# PSP — ALL CODING QUESTIONS, CHAPTER-WISE, WITH ANSWERS

Every PSP coding question ever captured (portal assignments + Contest 2 + lab), organised by IA-1 chapter L1–L12, Question → Answer format. Sources: portal export 23 Oct + live portal pages 28 Oct + Slack #psp-26/#psp-b1/#psp-b2. **100 coding questions.**

Answers for the 23 previously-unsolved questions were verified against their portal samples (53/53); all other answers were verified against the sample cases printed in each statement during authoring.

Companion MCQ book: `PSP–ALL–MCQ–QA.pdf`. Questions I could NOT fetch: final section of this file.

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# Chapter 1 — Flowcharts

No coded questions exist for L1 (flowcharts appeared only in the Contest-1 syllabus list).

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# Chapter 2 — Variables, Datatype, Typecasting, Print

## 2.1 Print Mirror Right Angle Triangle · [Assignment · SOLVED]

Print Mirror Right Angle Triangle

Write a Python program to print the mirror image of a right-angled triangle using asterisks (*). The triangle should have 5 rows. Each row should:

Contain stars in increasing order (1 in the first row, 2 in the second, and so on)

Include spaces on the left side so that the stars are right-aligned.

See the example below for clarity.

Input

You do not need to take any input.

Output

Print the mirror right-angled triangle as shown in the example.

Example

Output

```
    *
   *
  **
  **
```

***

***

****

****

*****

*****

**ANSWER**

No input needed: loop rows 1..5, print (5-i) spaces followed by i stars so the triangle is right-aligned. The scraped sample shows each row twice (portal rendering artifact); the statement says 5 rows, so the intended output is the 5-row mirror triangle.

```
for i in range(1, 6):

    print(' ' * (5 - i) + '*' * i)
```

## 2.2 Print Inverted Triangle · [Assignment · SOLVED]

Print Inverted Triangle

Write a Python program to print an inverted right-angled triangle using asterisks (*).

The triangle should have 5 rows.

The first row should contain 5 stars.

Each subsequent row should have one less star than the previous row.

Input

You do not need to take any input.

Output

Print the inverted right-angled triangle as shown in the example.

Example

Output:

*****

*****

****

****

***

***

**

**

*

*

**ANSWER**

No input; print 5,4,3,2,1 stars, one row per line (the duplicated rows in the statement example are a portal rendering artifact).

```
for i in range(5, 0, -1):

    print("*" * i)
```

## 2.3 Print Numbers Right-Angled Triangle · [Assignment · SOLVED]

Print Numbers Right-Angled Triangle

Write a Python program to print a right-angled triangle made of numbers.

The triangle should have 5 rows.

The first row should contain the number 1.

The second row should contain 1 2.

The third row should contain 1 2 3, and so on.

See the example below for clarity.

Note: There should be a space between each number in a row.

Input

You do not need to take any input.

Output

Print the right-angled triangle of numbers as shown in the example.

Example

Output

1

1

1

2

1 2

1

2

3

1 2 3

1

2

3

4

1 2 3 4

1

2

3

4

5

1 2 3 4 5

**ANSWER**

No input. The portal example block is garbled by text extraction, but the statement is clear: print rows 1..5 where row i is "1 2 ... i" space-separated, one row per line.

```
for i in range(1, 6):

    print(" ".join(str(j) for j in range(1, i + 1)))
```

## 2.4 Print Hollow Square · [Assignment · SOLVED]

Print Hollow Square

Write a Python program to print a hollow square pattern using asterisks (*).

The square should have 5 rows and 5 columns.

The stars should form the border of the square.

The inside of the square should be empty (filled with spaces).

See the example below for clarity.

Input

You do not need to take any input.

Output

Print the hollow square pattern as shown in the example.

Example

Output

*****

*****

*  *

*  *

*  *

*  *

*  *

*  *

*****

*****

**ANSWER**

No input needed: print a 5x5 square whose border rows are all stars and middle rows are star + 3 spaces + star. The scraped sample shows each row twice (portal rendering artifact); the statement says 5 rows, so the intended output is the 5-row hollow square.

```
for i in range(5):

    if i == 0 or i == 4:

        print('*****')

    else:

        print('*   *')
```

## 2.5 No Time to Newline · [Assignment · SOLVED]

No Time to Newline

Write a program to print the below quote in a single line.

Quote

1.Do not watch the clock

2.Do what it does keep going

Your task is to print only the line number 1 of the quote.

Line number 2 of the quote will automatically appear in the output section when you run your code.

Input

You do not need to take any input. You need to print only line number 1 of the Quote.

Output

The output will be the quote in a single line as per below example.

Example

Input:

Output:

Do not watch the clock Do what it does keep going

**ANSWER**

Print only quote line 1 with a trailing space (end=" "); the portal auto-appends line 2 to form the single output line.

```
print("Do not watch the clock", end=" ")
```

---

# Chapter 3 — Input & Arithmetic Operators

# 3.1 Swap Two Variables · [Assignment · SOLVED]

Swap Two Variables

2x

Write a program to swap two integers a and b, using a third variable.

Input

User Task

You need not handle input or output. Your task is to swap the numbers stored in variables a and b.

Custom Input

The first line of input contains an integer a.

The second and final line contains an integer b.

Output

You need not to output anything.

Custom Output

The driver code will print two space-separated numbers representing the swapped values.

Constraints

0

≤

$a$

≤

1

0

3

0≤a≤10

3

0

≤

$b$

≤

1

0

3

0≤b≤10

3

Example

Input

10

20

Output

20 10

Explanation

Given the two integer numbers 10 and 20, after the swap the numbers will be 20 and 10.

Hence, the output is 20 10.

**ANSWER**

The statement gives no function signature (only says a and b are stored and the driver prints them), so a self-contained program is provided: read a and b, swap via a third variable, print space-separated.

```
a = int(input())

b = int(input())

temp = a

a = b

b = temp

print(a, b)
```

## 3.2 Find The Square · [Assignment · SOLVED]

Find The Square

2x

Write a program to take a number X as input and print its square.

Input

The first and only line of input contains a single integer X.

Output

Print an integer, representing the square of X.

Constraints

0

≤

$X$

≤

5

0≤X≤5

The input number will always be an integer.

Example

Input

4

Output

16

Explanation

The given number is 4, and its square will be 4 * 4 = 16.

**ANSWER**

Read the integer and print its square.

```
X = int(input())

print(X * X)
```

## 3.3 Rectangle Area Calculator · [Assignment · SOLVED]

Rectangle Area Calculator

2x

Write a program that calculates the area of a rectangle given its length and width.

The program should prompt the user to enter the length and width of the rectangle, and then calculate and print the area.

Input

The first line of input contains a single integer, representing the length of rectangle.

The second line of input contains a single integer, representing the width of rectangle.

Output

Print the answer in following format:

"Area of the rectangle is: [calculated area]"

Constraints

1 ≤ $Length$, $Width$ ≤ 100

1≤Length,Width≤100

Length and width will be positive numbers.

Example

Input

5

3

Output

Area of the rectangle is: 15

**ANSWER**

Read length and width on separate lines; print area in the exact required format.

```
length = int(input())

width = int(input())

print(f"Area of the rectangle is: {length * width}")
```

## 3.4 Calculate and Print Age · [Assignment · SOLVED]

Calculate and Print Age

2x

Write a Python program that takes two integers as input: the birth year and the current year.

Your task is to calculate the age by subtracting the birth year from the current year, and print the result.

Input

The first line of input contains a single integer, the birth year.

The second line also contains a single integer, the current year.

Output

Print an integer, the calculated age.

Constraints

1950 <= Birth Year <= Current Year <= 2025

Example

Input

2000

2024

Output

24

Explanation

Brith Year = 2000

Current Year = 2024

Age = 2024 - 2000 = 24

Input

2006

2025

Output

19

Explanation

Brith Year = 2006

Current Year = 2025

Age = 2025 - 2006 = 19

**ANSWER**

Age is simply current year minus birth year.

```
birth_year = int(input())

current_year = int(input())

print(current_year - birth_year)
```

## 3.5 Energy Consumption Calculation · [Assignment · SOLVED]

Energy Consumption Calculation

2x

You are given two integers: units representing the total electricity consumed and cost representing the cost of each unit.

Your task is to write a Python program that calculates the total electricity cost and prints the result.

Input

The first and only line of input contains two space-separated integers, representing units and cost respectively.

Output

Print an integer representing the total electricity cost.

Constraints

0

≤

$u$

$n$

$i$

$t$

$s$

≤

1000

0≤units≤1000

1

≤

$c$

$o$

$s$

$t$

≤

1000

1≤cost≤1000

Example

Input

20 10

Output

200

Explanation

Since 20 units of electricity was consumed and the cost per unit is 10. The total cost would be 200.

**ANSWER**

Read units and cost, print their product.

```
units, cost = map(int, input().split())

print(units * cost)
```

## 3.6 Celsius to Fahrenheit Conversion · [Assignment · SOLVED]

Celsius to Fahrenheit Conversion - 2

2x

You are given an integer, representing the temperature in celsius.

Your task is to convert it to Fahrenheit, using the below given formula:

Fahrenheit = (Celsius × 9 / 5) + 32

You must complete the function celsius_to_fahrenheit(celsius) that returns the equivalent temperature in Fahrenheit.

Function Signature:

def celsius_to_fahrenheit(celsius):

It takes a single parameter, celsius, the temperature in Celsius.

Input

User Task

As this is a functional problem, there is no need to handle input from the user. Your task is to complete the function celsius_to_fahrenheit(celsius), which takes single parameter, celsius, the temperature in celsius.

Custom Input

The first and only line of input contains a single integer, celsius, representing the temperature in celsius.

Output

Return the corresponding temperature in Fahrenheit (as a float).

Constraints

-100 ≤ celsius ≤ 100

Example

Input

0

Output

32.0

Explanation

Input Celsius = 0

Fahrenheit = (0 × 9/5) + 32 = 0 + 32 = 32.0

So the output is 32.0.

Input

37

Output

98.6

Explanation

Input Celsius = 37

Fahrenheit = (37 × 9/5) + 32 = 98.6

So the output is 98.6.

**ANSWER**

Functional problem: return F = C * 9/5 + 32 as a float.

```
def celsius_to_fahrenheit(celsius):

    return celsius * 9 / 5 + 32
```

## 3.7 Convert Minute to Hour and Minute Format · [Assignment · SOLVED]

Convert Minute to Hour and Minute Format

2x

Write a program to convert total minutes into hours and remaining minutes.

Print the hours and minutes in the below given format:

Time: {x} hour(s) and {y} minute(s)

where {x} will be replaced by hours and {y} by minutes.

Input

The first and only line of input contains a single integer, representing the total minutes.

Output

Print the hours and minutes in the below given format:

Time: {x} hour(s) and {y} minute(s)

where {x} will be replaced by hours and {y} by minutes.

Constraints

0 ≤ total minutes ≤ 103

Example

Input

135

Output

Time: 2 hour(s) and 15 minute(s)

Input

60

Output

Time: 1 hour(s) and 0 minute(s)

**ANSWER**

Integer-divide by 60 for hours, remainder for minutes, and print in the exact required format.

```
total_minutes = int(input())

hours = total_minutes // 60

minutes = total_minutes % 60

print(f"Time: {hours} hour(s) and {minutes} minute(s)")
```

## 3.8 Average of Three · [Assignment · SOLVED]

Average of Three

2x

Write a program that takes three numbers as input and prints their average.

Note: The result should be rounded to two decimal places.

Input

The first and only line of input contains three space-separated numbers (integer or float).

Output

Print the average of the three numbers rounded to two decimal places.

Constraints

–

100

≤

$a$

,

$b$

,

$c$

≤

100

−100≤a,b,c≤100

Each number will be a valid integer or float.

Example

Input

10 20 30

Output

20.0

Explanation

Average = (10 + 20 + 30) / 3 = 60 / 3 = 20.0

Input

3.3 1.2 1

Output

1.83

Explanation

To calculate the average, we add all the values:

3.3 + 1.2 + 1 = 5.5

Now, divide the total by the number of values:

5.5 ÷ 3 = 1.8333333333333333

Since the question requires the answer to be rounded to two decimal places, we round 1.8333333333333333 to 1.83.

Final Answer: 1.83

**ANSWER**

Read the three numbers as floats, print round(avg, 2) so a whole average prints as 20.0 (not 20.00).

```
a, b, c = map(float, input().split())

print(round((a + b + c) / 3, 2))
```

## 3.9 Compound Interest Calculation in Python -1 · [Assignment · SOLVED]

Compound Interest Calculation in Python -1

2x

Write a program that calculates the compound interest.

The program should take principal amount (P), rate of interest (r), and time period (t) as inputs. Use the formula:

$C$
$I$
$=$
$($
$P$
$*$
$($
$1$
$+$
$($
$r$
$/$
$100$
$)$
$)$
$t$
$)$

–

$P$

CI=(P∗(1+(r/100))

t

)–P

Note: Display the result rounded to one decimal point.

Input

The first and only line of input contains three space-separated numbers, representing the principal amount (P), rate of interest (r), and time period (t) respectively

Note: The principal amount (P) and the time period (t) are provided as integers, while the rate of interest (r) is given as a floating-point number.

Output

The program should print the calculated compound interest.

Note: Display the result rounded to one decimal point.

Constraints

1

≤

$P$

≤

1

0

3

1≤P≤10

3

1

≤

$r$

≤

1

0

2

1≤r≤10

2

1

≤

$t$

≤

10

1≤t≤10

Example

Input:

1000 5.0 3

Output:

157.6

Explanation:

principal amount: 1000

rate of interest (as a percentage): 5

the time period (in years): 3

Compound interest: 157.63, calculated using formula CI = (P * (1 + (r / 100)) ^ t ) - P

**ANSWER**

Read P (int), r (float), t (int); CI = P*(1+r/100)**t - P; print rounded to 1 decimal.

```
data = input().split()

P = int(data[0])

r = float(data[1])

t = int(data[2])

ci = P * (1 + r / 100) ** t - P

print(round(ci, 1))
```

## 3.10 Celsius to Fahrenheit Conversion - 2 · [Assignment · SOLVED]

Celsius to Fahrenheit Conversion - 2

2x

You are given an integer, representing the temperature in celsius.

Your task is to convert it to Fahrenheit, using the below given formula:

Fahrenheit = (Celsius × 9 / 5) + 32

You must complete the function celsius_to_fahrenheit(celsius) that returns the equivalent temperature in Fahrenheit.

Function Signature:

def celsius_to_fahrenheit(celsius):

It takes a single parameter, celsius, the temperature in Celsius.

Input

User Task

As this is a functional problem, there is no need to handle input from the user. Your task is to complete the function celsius_to_fahrenheit(celsius), which takes single parameter, celsius, the temperature in celsius.

Custom Input

The first and only line of input contains a single integer, celsius, representing the temperature in celsius.

Output

Return the corresponding temperature in Fahrenheit (as a float).

Constraints

-100 ≤ celsius ≤ 100

Example

Input

0

Output

32.0

Explanation

Input Celsius = 0

Fahrenheit = (0 × 9/5) + 32 = 0 + 32 = 32.0

So the output is 32.0.

Input

37

Output

98.6

Explanation

Input Celsius = 37

Fahrenheit = (37 × 9/5) + 32 = 98.6

So the output is 98.6.

**ANSWER**

Functional problem (same task as 011): return C*9/5+32 as a float.

```
def celsius_to_fahrenheit(celsius):

    return celsius * 9 / 5 + 32
```

## 3.11 Find the remainder · [Contest 2 · SOLVED]

Find the remainder

2x

You are given two numbers a and b. Find the remainder when a is divided by b.

Input

The first line contains integer a.

The second line contains integer b.

Output

Print the remainder when a is divided by b.

Constraints

0

≤

$a$

≤

200

0≤a≤200

1

≤

$b$

≤

200

1≤b≤200

Example

Input

9

7

Output

2

Explanation

When we divide 9 by 7, we get 2 as remainder.

Therefore, ans = 2.

**ANSWER**

Use Python's % operator; b is guaranteed ≥ 1.

```
a = int(input())

b = int(input())

print(a % b)
```

## 3.12 Newton and Rounding · [Contest 2 · was unsolved — solved here]

Newton and Rounding

Newton has a non-negative integer n. He wants to round it to the nearest multiple of 10.

If n already ends with 0, it is already rounded.

If the last digit is 1–4, move to the lower multiple of 10.

If the last digit is 6–9, move to the higher multiple of 10.

If the last digit is 5, always move to the lower multiple of 10.

Your task is to find and print the number Newton will round to.

Input

A single integer n — the number that Newton has.

Output

Print the result of rounding n.

Constraints

0

≤

$n$

≤

5000

0≤n≤5000

Example

Input

5

Output

0

Explanation

Normally, 5 could be rounded to 0 or 10, but as per the problem rule, when n = 5, always output 0.

Input

113

Output

110

Explanation

113 is closer to 110 than 120, so the rounded number is 110.

Input

2359

Output

2360

Explanation

2359 is closer to 2360 than 2350, so it rounds up.

**ANSWER**

(Solution verified against portal samples — 53/53 pass across the full set.)

```
n = int(input())

d = n % 10

if d <= 5:

    print(n - d)        # 0,1,2,3,4,5 all go DOWN (5 is the trap)

else:

    print(n + (10 - d))
```

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# Chapter 4 — if-else

## 4.1 Even or Odd Checker · [Assignment · SOLVED]

Even or Odd Checker

2x

Write a Python program that checks if a given number is even or odd.

Input

The input contains a single integer number

Output

The program should print a single word either - Even or Odd according to number.

Example

Input:

7

Output:

Odd

Input:

6

Output:

Even

**ANSWER**

Check divisibility by 2 with the modulo operator.

```
n = int(input())

if n % 2 == 0:

    print("Even")

else:

    print("Odd")
```

## 4.2 Integer Type · [Assignment · SOLVED]

Integer Type

2x

You are given a non-zero integer num. Your task is to check whether the number is Positive or Negative.

Input

The first and only line contains an integer num.

Output

Print exactly one word, either Positive or Negative, depending on the value of the input.

Constraints

–

30

≤

$n$

$u$

$m$

≤

30

−30≤num≤30

$n$

$u$

$m$

≠

0

num

□

=0

Example

Input:

5

Output:

Positive

Input:

-9

Output:

Negative

**ANSWER**

The number is non-zero: print Positive if > 0, else Negative.

```
num = int(input())

if num > 0:

    print("Positive")

else:

    print("Negative")
```

## 4.3 Who is Greater? · [Assignment · SOLVED]

Who is Greater?

2x

You are given two distinct integers a and b. Your task is to compare two numbers a and b, and print the comparison in the following format:

print "a is greater than b",  if a is greater than b, or

print "b is greater than a", if b is greater than a

Note: You are guaranteed that a ≠ b.

Input

A single line containing two space-separated integers a and b.

Output

Print exactly one line in one of the following formats:

a is greater than b — if a is greater than b

b is greater than a — if b is greater than a

Note: You must use the actual values of a and b in the output.

Constraints

-102 ≤ a, b ≤ 102

a ≠ b

Example

Input:

12 5

Output:

12 is greater than 5

Input:

-3 10

Output:

10 is greater than -3

**ANSWER**

Read a, b (guaranteed distinct); print which is greater using actual values.

```
a, b = map(int, input().split())

if a > b:

    print(f"{a} is greater than {b}")

else:

    print(f"{b} is greater than {a}")
```

## 4.4 Weather Report · [Assignment · SOLVED]

Weather Report

2x

Write a program that takes an integer current temperature as input, and determines the type of weather using the below conditions:

Temp < 0 : Freezing Weather

Temp 0-9 : Very Cold

Temp 10-19 : Cold

Temp 20-29 : Normal

Temp 30-39 :  Hot

Temp >=40 : Very Hot

Input

A line containing an integer number representing the temperature.

Output

Print the weather according to the temperature as The weather today is classified as: answer

Example

Input

29

Output

The weather today is classified as: Normal

Explanation

According to the given question the temperature 20-29 are considered Normal

Input

9

Output

The weather today is classified as: Very Cold

Explanation

According to the given question the temperature 0-9 are considered Very Cold

**ANSWER**

If-elif ladder over the given temperature bands (30-39 is "Hot"; the double space in the statement is an extraction artifact).

```
temp = int(input())

if temp < 0:

    ans = "Freezing Weather"

elif temp <= 9:

    ans = "Very Cold"

elif temp <= 19:

    ans = "Cold"

elif temp <= 29:

    ans = "Normal"

elif temp <= 39:

    ans = "Hot"

else:

    ans = "Very Hot"

print("The weather today is classified as:", ans)
```

## 4.5 AM_PM · [Assignment · SOLVED]

AM_PM

2x

Write a program that takes the hour (0-23) in 24-hour format as input and prints whether it is AM or PM.

Note:

The time is considered AM if the hour is between 0 and 11 (inclusive).
The time is considered PM if the hour is between 12 and 23 (inclusive).
Input
First line will contain an integer hour (0-23) as input.
Output
The program should print the appropriate answer(AM or PM) based on the hour.
Constraints
0 ≤ hour ≤ 23
Example
Example 1
Input:
10

Output:
AM

Example 2

Input:

13

Output:

PM

**ANSWER**

Hours 0-11 are AM, 12-23 are PM.

```
hour = int(input())

if hour <= 11:

    print("AM")

else:

    print("PM")
```

## 4.6 Identify Square or Rectangle · [Assignment · SOLVED]

Identify Square or Rectangle

2x

You are given two integers representing the length and breadth of a rectangle.

You need to print "Square" if the given rectangle is square as well, otherwise print "Rectangle".

Input

The first and only line of input contains two space-separated integers, representing the length and breadth of the rectangle respectively.

Output

Print "Square" or "Rectangle" based on the dimensions.

Constraints

1

≤

$l$

$e$

$n$

$g$

$t$

$h$

,

$b$

$r$

$e$

$a$

$d$

$t$

$h$

≤

100

1≤length,breadth≤100

Example

Input

5 5

Output

Square

Input

10 9

Output

Rectangle

**ANSWER**

Print Square when length equals breadth, else Rectangle.

```
length, breadth = map(int, input().split())

if length == breadth:

    print("Square")

else:

    print("Rectangle")
```

## 4.7 Absolute Difference of Two Numbers · [Assignment · SOLVED]

Absolute Difference of Two Numbers

2x

Rakshit has two numbers, and he wants to find how far apart they are on the number line — that is, their absolute difference.

Your task is to find and print the absolute difference of the two given numbers.

Input

The first line of input contains two space-separated integers a and b.

Output

Print the absolute difference of a and b in the output.

Constraints

1

≤

$a$

,

$b$

≤

1000

1≤a,b≤1000

Example

Input

8 19

Output

11

Explanation

The absolute difference between 8 and 19 is | 8 - 19 | = 11.

**ANSWER**

Use the built-in abs() on the difference of the two integers read from one line.

```
a, b = map(int, input().split())

print(abs(a - b))
```

## 4.8 Check if Profit or Loss · [Assignment · SOLVED]

Check if Profit or Loss

2x

You are given two integers:

CP — the Cost Price of an item

SP — the Selling Price of the item

Your task is to determine whether the transaction resulted in a Profit or a Loss.

Note: There will be no test case where SP = CP.

Input

The first line contains two space-separated integers CP and SP.

Output

Print:

Profit if the selling price is greater than the cost price.

Loss if the selling price is less than the cost price.

Constraints

1 ≤ CP, SP ≤ 109

Example

Input

100 150

Output

Profit

Explanation

Since SP > CP, the transaction resulted in a profit.

Input

200 120

Output

Loss

Explanation

Since SP < CP, the transaction resulted in a loss.

**ANSWER**

SP > CP means Profit, otherwise Loss (SP != CP is guaranteed).

```
CP, SP = map(int, input().split())

if SP > CP:

    print("Profit")

else:

    print("Loss")
```

## 4.9 Voter Eligibility II · [Assignment · SOLVED]

Voter Eligibility II

2x

You are given an integer input, representing the age of a person.

You task is to check whether the person is eligible to vote or not.

A person is eligible to vote if their age is greater than or equal to 18.

Print Yes if the person can vote, otherwise print No.

Input

The first and only line of input contains a single integer, representing the age.

Output

Print Yes if the person can vote, otherwise print No.

Constraints

0

≤

$a$

$g$

$e$

≤

100

0≤age≤100

Example

Input

20

Output

Yes

Input

15

Output

No

**ANSWER**

Eligible when age >= 18.

```
age = int(input())

if age >= 18:

    print("Yes")

else:

    print("No")
```

## 4.10 Check Profit or Loss · [Assignment · SOLVED]

Check Profit or Loss

You are given two integers, cost and sell, representing the cost price and the selling price of a product.

Your task is to determine whether the transaction results in a profit, a loss, or no profit and no loss.

Also, you need to calculate the corresponding amount.

Input

The first line of input contains a single integer, representing cost.

The second line also contains a single integer, representing sell.

Output

The program should print two lines.

The first line should contain one of the words Profit, Loss, or No Profit No Loss.

The second line should contain an integer representing the profit or loss amount, and it should be 0 in case of no profit and no loss.

Constraints

1

≤

$c$

$o$

$s$

$t$

≤

1000

1≤cost≤1000

1

≤

$s$

$e$

$l$

$l$

≤

1000

1≤sell≤1000

Example

Input

150

100

Output

Loss

50

Input

100

100

Output

No Profit No Loss

0

**ANSWER**

Two-line input (cost, sell); print status line then amount line.

```
cost = int(input())

sell = int(input())

if sell > cost:

    print("Profit")

    print(sell – cost)

elif sell < cost:

    print("Loss")

    print(cost – sell)

else:

    print("No Profit No Loss")

    print(0)
```

# 4.11 The Frame · [Contest 2 · SOLVED]

The Frame

2x

In the annual design festival of a small town, a young engineer is given four wooden pieces to create the frame of a special display panel.

Each piece is straight and has a fixed length. The engineer is not allowed to cut, break or bend any of the pieces. All four pieces must be used exactly once.

The display panel is supposed to have a perfectly balanced shape. This means that its four sides must have exactly the same length, and the four pieces must be arranged so that they form a proper square.

The engineer has received the lengths of the four available pieces. Determine whether the four pieces can be arranged to form the required shape.

Input

The input contains a single line with four integers a, b, c, and d, representing the lengths of the four wooden pieces.

1 ≤ a, b, c, d ≤ 10

All four pieces must be used.

No piece can be cut, bent, broken, or modified.

Output

Print YES if the four pieces can be arranged to form a square.

Otherwise, print NO.

Constraints

1

≤

$a$

,

$b$

,

$c$

,

$d$

≤

10

1≤a,b,c,d≤10

All four pieces must be used exactly once.

The pieces cannot be modified in any way.

Example

Example 1

Input:

1 2 3 4

Output:

NO

Explanation:

The four pieces have different lengths, so they cannot form desired shape.

Example 2

Input:

5 5 5 5

Output:

YES

Explanation:

All four pieces have the same length. Therefore, they can be arranged as the desired shape.

**ANSWER**

Four uncut pieces form a square iff all four lengths are equal; print YES/NO accordingly.

```
a, b, c, d = map(int, input().split())

if a == b == c == d:

    print("YES")

else:

    print("NO")
```

---

# Chapter 5 — if-elif-else ladder

## 5.1 Grade Calculator · [Assignment · SOLVED]

Grade Calculator

Write a Python program that takes a student's score as input and prints their grade based on the following criteria:

Grading Criteria:

If the score is 90 or above, the grade is A.

If the score is 80 or more but less than 90, the grade is B.

If the score is 70 or more but less than 80, the grade is C.

If the score is 60 or more but less than 70, the grade is D.

If the score is less than 60, the grade is F.

Input

The one and only line containing a single integer representing the student's score.

Output

Print in a line containing the grade corresponding to the score.

Constraints

1<=score<=100

Example

Sample 1:

Input:

85

Output:

Grade B

Explanation:

The score is 85, which falls in the range of 80 or more but less than 90. So, the grade is B.

Sample 2:

Input:

60

Output:

Grade D

Explanation:

The score is 60, which is 60 or more but less than 70. Therefore, the grade is D.

**ANSWER**

Top-down if-elif ladder on the score thresholds; output is "Grade" plus the letter.

```
score = int(input())
if score >= 90:
    print("Grade A")
elif score >= 80:
    print("Grade B")
elif score >= 70:
    print("Grade C")
elif score >= 60:
    print("Grade D")
else:
    print("Grade F")
```

## 5.2 The Number Detective · [Assignment · SOLVED]

The Number Detective

Given a number n, write a Python program to check whether the number n is positive, negative, or zero.

Print Positive if n is greater than 0

Print Negative if n is less than 0

Print Zero if n is equal to 0

Input

The one and only line contains a single integer n.

Output

Print whether the number is Positive, Negative, or Zero.

Constraints

-1000 ≤ number ≤ 1000

Example

Sample Input 1:

13

Sample Output 1:

Positive

Explanation:

Since 13 is greater than 0, that's why the output is Positive number.

Sample Input 2:

-13

Sample Output 2:

Negative

Explanation:

Since -13 is less than 0, that's why the output is Negative number.

**ANSWER**

If-elif-else on the sign of n: Positive, Negative, or Zero.

```
n = int(input())

if n > 0:

    print("Positive")

elif n < 0:

    print("Negative")

else:

    print("Zero")
```

## 5.3 Greater or Equal · [Assignment · SOLVED]

Greater or Equal

You are given two integers a and b.

Your task is to compare the two numbers and print:

a is greater than b if the value of a is greater than the value of b

b is greater than a if the value of b is greater than the value of a

a and b are equal if both values are equal

Note: Replace a and b with their actual input values in the output.

Input

The first line contains two space-separated integers a and b.

Output

Print exactly one line in one of the following formats:

a is greater than b — if the value of a is greater than the value of b

b is greater than a — if the value of b is greater than the value of a

a and b are equal — if both values are equal

Note: Replace a and b with their actual input values in the output.

Constraints

1 ≤ $a$ , $b$ ≤ 100

1≤a,b≤100

Example

Example 1

Input

12 5

Output

12 is greater than 5

Explanation

Among the two given numbers, 12 is greater than 5.

Example 2

Input

1 10

Output

10 is greater than 1

Explanation

Among the two given numbers, 10 is greater than 1.

Example 3

Input

7 7

Output

7 and 7 are equal

Explanation

Both numbers have the same value.

**ANSWER**

Three-way comparison with actual values in the output.

```
a, b = map(int, input().split())

if a > b:

    print(f"{a} is greater than {b}")

elif b > a:

    print(f"{b} is greater than {a}")

else:

    print(f"{a} and {b} are equal")
```

## 5.4 FizzBuzz II · [Assignment · SOLVED]

FizzBuzz II

2x

Arjun has recently started learning about divisibility rules. His friend gave him a simple problem to test his understanding.

Given an integer n, your task is to help Arjun print:

"FizzBuzz" if n is divisible by both 3 and 5.

"Fizz" if n is divisible by 3 only.

"Buzz" if n is divisible by 5 only.

Otherwise, print the number n itself.

Input

A single integer n — the number to be evaluated.

Output

Print:

"FizzBuzz" if divisible by both 3 and 5

"Fizz" if divisible only by 3

"Buzz" if divisible only by 5

Else, print the number itself

Constraints

-10^9 <= n <= 10^9

Example

1. Sample Input:

15

Sample Output:

FizzBuzz

Explanation: 15 is divisible by 3 and 5, hence the output as per the condition is "FizzBuzz".

2. Sample Input:

5

Sample Output:

Buzz

Explanation: 5 is divisible by 5, hence the output as per the condition is "Buzz".

3. Sample Input:

3

Sample Output:

Fizz

Explanation: 3 is divisible by 3, hence the output as per the condition is "Fizz".

**ANSWER**

Test divisibility by 15 first, then 3, then 5; otherwise print the number. Works for negatives too since Python's % is sign-consistent for divisibility.

```
n = int(input())

if n % 15 == 0:

    print("FizzBuzz")

elif n % 3 == 0:

    print("Fizz")

elif n % 5 == 0:

    print("Buzz")

else:

    print(n)
```

## 5.5 Income Tax Calculator · [Assignment · SOLVED]

Income Tax Calculator

2x

Write a program to calculate the income tax for an individual based on their annual income in Indian Rupees (₹) provided as input. The tax is calculated according to the following slabs:

Income Range (₹) Tax Calculation

Up to ₹2,50,000 No tax.

₹2,50,001 to ₹5,00,000 5% tax on amount between ₹2,50,001 to ₹5,00,000

₹5,00,001 to ₹10,00,000 ₹12,500 plus 10% tax on amount between ₹5,00,001 to ₹10,00,000

Above ₹10,00,000 ₹62,500 plus 20% tax on amount above ₹10,00,000.

The program should read the annual income from the user, calculate the tax accordingly, and output the tax amount upto 1 decimal places.

Note: We can round the output to 1 decimal place using print(round(tax,1)).

Input

A single integer representing the annual income in rupees.

Output

Print the tax amount as a float number up to 1 decimal places.

Constraints

1

≤

Income

≤

1

0

9

1≤Income≤10

9

Example

input

200000

output

0.0

Explanation: As income is less than 250000, there will not be any tax that the individual is paying, that's why the output is 0.0.

input

550000

output

17500.0

Explanation:

Total income = 550000

1. No tax upto 250000

2. 5% tax in between 250000 to 500000 = 250000*0.05 = 12500.0

3. 10% tax in between 500000 to 1000000,

there for remaining taxable income = 550000 - 500000 = 50000

10% of 50000 = 50000 * 0.1 = 5000.0

Total Tax to be paid = 0.0 + 12500.0 + 5000.0 = 17500.0

**ANSWER**

Apply the slab ladder: 0 below 2.5L, 5% on (income-2.5L) up to 5L, 12500 + 10% on (income-5L) up to 10L, 62500 + 20% above 10L; print rounded to 1 decimal.

```
income = int(input())

if income > 1000000:

    tax = 62500 + 0.2 * (income - 1000000)

elif income > 500000:

    tax = 12500 + 0.1 * (income - 500000)

elif income > 250000:

    tax = 0.05 * (income - 250000)

else:

    tax = 0.0

print(round(tax, 1))
```

## 5.6 Calculate Electricity Bill · [Assignment · SOLVED]

Calculate Electricity Bill - 2

2x

An electricity board calculates monthly power usage bills based on the number of units consumed using a slab-wise tariff system. Each slab charges a different rate per unit as follows:

First 100 units: ₹4 per unit

Next 100 units (i.e., units 101–200): ₹5 per unit

Next 100 units (i.e., units 201–300): ₹6 per unit

Next 100 units (i.e., units 301–400): ₹7 per unit

Units above 400: ₹8 per unit

After calculating the base bill, a 10% surcharge is added to the total bill.

You are required to implement the function calculate_bill(units) that calculates the total electricity bill including the surcharge, and returns a float rounded to 1 decimal place using the built-in round() function.

Function Signature:

def calculate_bill(units): which takes single parameter: units: represents the number of electricity units consumed.

Input

User Task

As this is a functional problem, there is no need to handle input from the user. Your task is to complete the function calculate_bill(units), which takes single parameter: units: represents the number of electricity units consumed.

Custom Input

A Single line containing an integer representing the number of units.

Output

Return a float — the total electricity bill including the 10% surcharge, rounded to 1 decimal place using round().

Example

Input

1

Output

Electricity Bill: Rs. 4.4

Explanation

Units are less than 100 therefore total cost is unit charge + surcharge 4 X 1 + 4 X  0.1 = 4.4

Input

100

Output

Electricity Bill: Rs. 440.0

Explanation

Units are equal to 100 therefore total cost is unit charge + surcharge 4 X 100 + 400 X  0.1 = 440.0

Input

340

Output

Electricity Bill: Rs. 1958.0

Explanation

For the first 100 units(i.e from 1 to 100) cost will be: 100*4 = 400

For next 100 units(i.e from 101 to 200) cost will be: 100*5 = 500

For next 100 units(i.e from 201 to 300) cost will be: 100*6 = 600

For remaining 40 units(i.e from 301 to 340) cost will be: 40*7 = 280

Total Cost excluding surcharge = 400 + 500 + 600 + 280 = 1780

Final Bill including 10% surcharge on Total cost = 1780 + (1780*0.1) = 1780 + 178 = 1958

**ANSWER**

Functional problem: slab-wise base bill (4/5/6/7/8 per unit), then +10% surcharge; return round(total, 1).

```
def calculate_bill(units):

    bill = min(units, 100) * 4

    if units > 100:

        bill += min(units - 100, 100) * 5

    if units > 200:

        bill += min(units - 200, 100) * 6

    if units > 300:

        bill += min(units - 300, 100) * 7

    if units > 400:

        bill += (units - 400) * 8

    return round(bill * 1.1, 1)
```

# 5.7 Triangle Type Checker · [Assignment · SOLVED]

Triangle Type Checker

2x

You are given the lengths of the three sides of a triangle. Your task is to determine and print the type of triangle formed.

Use the following rules:

Equilateral: All three sides are equal

Isosceles: Exactly two sides are equal

Scalene: All sides are different

Input

A single line contains three space-separated integers representing the lengths of the three sides of a triangle.

Output

Print one of the following based on the type of triangle:

Equilateral

Isosceles

Scalene

Note: Your output must match the cases exactly as shown above. Outputs are case-sensitive.

Constraints

$1 \leq a, b, c \leq 100$

The input will always represent a valid triangle.

Example

input

3 4 3

output

Isosceles

Explanation:

Two sides of a triangle are equal -> Isosceles

**ANSWER**

Classify by equality of the three sides; output case must match exactly (Equilateral/Isosceles/Scalene).

```
a, b, c = map(int, input().split())

if a == b == c:

    print("Equilateral")

elif a == b or b == c or a == c:

    print("Isosceles")

else:

    print("Scalene")
```

## 5.8 Check if All Three Numbers are Equal · [Assignment · SOLVED]

Check if All Three Numbers are Equal

2x

You are given three integers a, b, and c.

Your task is to determine whether all three numbers are equal.

Print:

Numbers are equal if all three numbers have the same value.
Numbers are not equal otherwise.
Input
The first line contains three space-separated integers a, b, and c.
Output

Print exactly one line:

Numbers are equal if all three numbers are equal.
Numbers are not equal otherwise.
Constraints
1
≤
$a$
,
$b$
,
$c$
≤
100
1≤a,b,c≤100
Example
Example 1
Input
5 5 5

Output

Numbers are equal

Explanation

All three numbers have the same value.

Example 2

Input

5 5 7

Output

Numbers are not equal

Explanation

The three numbers do not all have the same value.

**ANSWER**

Print "Numbers are equal" only when a == b == c, else "Numbers are not equal".

```
a, b, c = map(int, input().split())

if a == b == c:

    print("Numbers are equal")

else:

    print("Numbers are not equal")
```

## 5.9 Find the Biggest ! · [Assignment · SOLVED]

Find the Biggest !

2x

Comparing numbers is a common task in programming. Your goal is to determine the largest of three given numbers.

Your task:

Given three integers A, B, and C, print the largest among them.

Input

The first and only line of input contains three space-separated integers, A, B, and C respectively.

Output

Print an integer, the largest number among the given three numbers.

Constraints

0

≤

$A$

,

$B$

,

$C$

≤

100

0≤A,B,C≤100

Example

Input

1 2 3

Output

3

Input

5 3 1

Output

5

Input

10 9 10

Output

10

**ANSWER**

If-elif ladder over three integers.

```
A, B, C = map(int, input().split())

if A >= B and A >= C:

    print(A)

elif B >= A and B >= C:

    print(B)

else:

    print(C)
```

## 5.10 Library fine · [Assignment · SOLVED]

Library fine

Calculate the fine for a late return of a library book based on the following details:

For the first 3 days, the fine is Rs. 10.

For the next 4 to 10 days, the fine is Rs. 20.

For any period exceeding 10 days, the fine is Rs.100.

Input

A line containing an integer representing the number of late days.

Output

Print a line containing the total fine amount.

Example

Input

10

Output

20

Explanation

Fine for 10 late days is 20 Rs according to the question.

Input

20

Output

100

Explanation

Fine for 10 late days is 100 Rs according to the question.

**ANSWER**

Flat fine per band: 1-3 days → 10, 4-10 days → 20, more than 10 → 100 (statement gives no rule for 0 days; the ladder treats anything ≤ 3 as 10).

```
days = int(input())

if days <= 3:

    print(10)

elif days <= 10:

    print(20)

else:

    print(100)
```

## 5.11 Calculate salary · [Assignment · SOLVED]

Calculate salary

Calculate the gross salary of an employee based on their basic salary according to the following criteria:

basic salary <= 10,000, HRA = 20% and DA = 80%.

20,000 >= basic salaries > 10,000 , the HRA = 25% and DA = 90%.

basic salary > 20,000, HRA = 30% and DA = 95%.

Hint: Gross Salary = Basic Salary + HRA + DA

Input

A line containing an integer representing the basic salary of the employee.

Output

Print the total gross salary as The gross salary of the employee is: answer

Example

Input

10001

Output

The gross salary of the employee is: 21502.15

Explanation

By using the formula answer is 21502.15

**ANSWER**

Pick HRA/DA rates by basic slab and add to basic. Compute the total in integer paisa (HRA/DA percentages map exactly to paisa multiples of the rupee amount) then divide by 100 once, avoiding float artifacts and printing the exact sample value 21502.15.

```
basic = int(input())

if basic <= 10000:

    paisa = basic * 100 + basic * 20 + basic * 80

elif basic <= 20000:

    paisa = basic * 100 + basic * 25 + basic * 90

else:

    paisa = basic * 100 + basic * 30 + basic * 95

print("The gross salary of the employee is:", paisa / 100)
```

## 5.12 BMI Calculator-1 · [Assignment · SOLVED]

BMI Calculator-1

You are asked to determine a person's Body Mass Index (BMI) based on their weight and height.

The BMI is calculated using the following formula:

BMI = weight / (height_in_meters  x  height_in_meters)

Here,

weight is given in kilograms
height is given in centimeters and must be converted to meters

You must round the BMI value to exactly 1 decimal place using the built-in round() function.

Based on the rounded BMI value, print the corresponding BMI category:

If BMI is less than 18.5, print Underweight
If BMI is between 18.5 and 24.9 (inclusive), print Normal weight
If BMI is between 25 and 29.9 (inclusive), print Overweight
If BMI is 30 or greater, print Obese
Input
The input consists of two lines as follows:
The first line contains an integer weight — the person's weight in kilograms.

The second line contains an integer height — the person's height in centimeters.

Output

Print two lines:

First line: the BMI value rounded to one decimal place using round()

Second line: the corresponding BMI category.

Constraints

1 ≤ weight ≤ 300

50 ≤ height ≤ 250

Example

Input

60

165

Output

22.0

Normal weight

Explanation:

BMI = 60/1.65*1.65 = 22.0

22.0 is in range 18.5 to 24.9

Category = Normal weight

**ANSWER**

Convert height cm→m, compute weight/(h*h), round to 1 decimal (prints as "22.0"), then classify the rounded value.

```
weight = int(input())

height = int(input())

h = height / 100

bmi = round(weight / (h * h), 1)

print(bmi)

if bmi < 18.5:

    print("Underweight")

elif bmi < 25:

    print("Normal weight")

elif bmi < 30:

    print("Overweight")

else:

    print("Obese")
```

---

# Chapter 6 — Nested if-else

## 6.1 Can It Be a Triangle ? · [Assignment · SOLVED]

Can It Be a Triangle ?

You are given the lengths of three sides. Your task is to check whether these three sides can form a valid triangle.

Input

The first and only line of input contains three space-separated integers, representing the lengths of three sides.

Output

Print Yes if the sides form a valid triangle.

Print No otherwise.

Constraints

1

≤

$S$

$i$

$d$

$e$

$L$

$e$

$n$

$g$

$t$

$h$

≤

100

1≤SideLength≤100

Example

Input

20 10 13

Output

Yes

Explanation

All combinations of side pairs sum to more than the third side. So, a triangle is possible.

Input

2 5 3

Output

No

Explanation

2 + 3 is not greater than 5, so a triangle cannot be formed.

**ANSWER**

Valid triangle iff every pair sum is strictly greater than the third side.

```
a, b, c = map(int, input().split())

if a + b > c and a + c > b and b + c > a:

    print("Yes")

else:

    print("No")
```

## 6.2 Leap Year Checker · [Assignment · SOLVED]

Leap Year Checker

2x

You are given an integer representing a year. Your task is to determine whether it is a leap year or not.

Input

The input contains a single integer: year — the year to check

Output

Print a single line as follows:

Leap year if the year is a leap year

Not a leap year otherwise

Constraints

1 ≤ year ≤ 10^9

Example

Example 1:

Input:

2000

Output:

Leap year

Explanation:

2000 is divisible by 400 ⇒

2000

mod

400

=

0

2000mod400=0

So, it's a leap year.

Example 2:

Input:

1900

Output:

Not a leap year

Explanation:

1900 is divisible by 100 ⇒

1900

mod

100

=

0

1900mod100=0

but not divisible by 400 ⇒

1900

mod

400

≠

0

1900mod400

□

=0

So, it's not a leap year.

**ANSWER**

Leap iff divisible by 400, or by 4 and not by 100.

```
year = int(input())

if year % 400 == 0 or (year % 4 == 0 and year % 100 != 0):

    print("Leap year")

else:

    print("Not a leap year")
```

## 6.3 Roller Coaster Ride · [Assignment · SOLVED]

Roller Coaster Ride

A theme park wants to check if Alice can ride their roller coaster. You are given the height (in cm) and age (in years) of Alice.

Eligibility and ticket rules are as follows:

Height Requirements:

Less than 120 cm → Not Allowed

120 – 140 cm → Must ride with Adult

Above 140 cm → Can ride alone

Age Restrictions:

Less than 5 years → Not Allowed

5 – 12 years → Must ride with Adult

Above 12 → Can ride alone

Ticket Price Calculation:

If Allowed Alone → ₹300

Allowed with Adult → ₹200

Your task is to print whether Alice is allowed to ride or not. If Alice is allowed to ride, print the ticket price of ride in the next line.

Note: Alice's eligibility is determined by both height and age. If either height or age imposes a stricter restriction, that restriction takes precedence.

For example, if her age requires her to ride with an adult, she cannot ride alone even if her height allows it.

Similarly, if her age or height disqualifies her from riding, she is not allowed to ride regardless of the other factor.

Input

The first line of input contains a single integer, representing height.

The second line also contains a single integer, representing age.

Output

The output should contain one or two lines:

On the first line, print Alice's riding status:

Not Allowed — if Alice is not allowed to ride.

Must ride with Adult — if Alice can ride only with an adult.

Can ride alone — if Alice can ride by herself.

If Alice is allowed to ride (either with an adult or alone), print the ticket price on the second line (without ₹ sign).

Constraints

50

≤

$h$

$e$

$i$

$g$

$h$

$t$

≤

300

50≤height≤300

1

≤

$a$

$g$

$e$

≤

100

1≤age≤100

Example

Input

145

14

Output

Can ride alone

300

Explanation

Height of Alice is greater than 140 cm and also the age is above 12.

Therefore, Alice can ride alone and the ticket price will be ₹300.

Input

110

4

Output

Not Allowed

**ANSWER**

Map height and age each to a level (0=not allowed, 1=with adult, 2=alone); the stricter (lower) level wins. Print price on the second line only when allowed.

```
height = int(input())

age = int(input())

if height < 120 or age < 5:

    print("Not Allowed")

elif height <= 140 or age <= 12:

    print("Must ride with Adult")

    print(200)

else:

    print("Can ride alone")

    print(300)
```

## 6.4 Accommodation Decision · [Assignment · SOLVED]

Accommodation Decision

A student wants to check whether they will stay On-Campus or Off-Campus based on the rules provided in the NST rulebook.

The rules for accommodation are:

If the student is a girl, she will definitely stay On-Campus, irrespective of the year.

If the student is a boy:

In first year → he will stay On-Campus

In second year → he will stay Off-Campus

Given the student's gender and year of study, determine their accommodation type.

Input

The input consists of two lines:

The first line contains a string gender, representing the gender of the student:

G for girl

B for boy

The second line contains an integer year, representing the year of study:

1 for first year

2 for second year

Output

Print a single line, either:

On-Campus or Off-Campus, according to the rules.

Constraints

The value of gender will be either "G" or "B".

The value of year will be either 1 or 2.

Example

Input

G

2

Output

On-Campus

Input

B

2

Output

Off-Campus

**ANSWER**

Girls are always On-Campus; boys are On-Campus only in year 1, else Off-Campus.

```
gender = input().strip()

year = int(input())

if gender == "G":

    print("On-Campus")

else:

    if year == 1:

        print("On-Campus")

    else:

        print("Off-Campus")
```

## 6.5 Calculate Tax Based on Age and Income · [Assignment · SOLVED]

Calculate Tax Based on Age and Income

You are given two values:

age — the age of a person
income — the annual income of the person (in rupees)

Your task is to calculate the amount of tax that the person needs to pay.

The tax rate depends on both the person's age and annual income.

Tax Slabs
Income Bracket Tax Rate (Age < 18) Tax Rate (Age ≥ 18)
Income ≤ 10,00,000 5% 10%
10,00,000 < Income ≤ 30,00,000 15% 20%
Income > 30,00,000 25% 30%

Print the tax amount that must be paid.

Input

The first line contains two space-separated integers age and income.

Output

Print a single integer representing the tax amount.

Constraints
1 ≤ age ≤ 120
0 ≤ income ≤ 109
Example
Example 1
Input
25 800000

Output

80000

Explanation The person's age is at least 18 and the income falls in the first tax slab. Therefore, the applicable tax rate is 10%. Tax = 10% of 800000 = 80000.

Example 2

Input

15 2000000

Output

300000

Explanation The person's age is less than 18 and the income falls in the second tax slab. Therefore, the applicable tax rate is 15%. Tax = 15% of 2000000 = 300000.

Example 3

Input

40 5000000

Output

1500000

Explanation The person's age is at least 18 and the income falls in the third tax slab. Therefore, the applicable tax rate is 30%. Tax = 30% of 5000000 = 1500000.

**ANSWER**

Pick rate from the age/income slab table; print integer tax = income*rate//100.

```
age, income = map(int, input().split())

if income <= 1000000:

    rate = 5 if age < 18 else 10

elif income <= 3000000:

    rate = 15 if age < 18 else 20

else:

    rate = 25 if age < 18 else 30

print(income * rate // 100)
```

## 6.6 Find the Smallest ! · [Assignment · SOLVED]

Find the Smallest !

Just like finding the largest value, identifying the smallest number is another common task in programming.

Your task:

Given three integers A, B, and C, print the smallest among them.

Input

The first and only line of input contains three space-separated integers, A, B, and C respectively.

Output

Print an integer, the smallest number among the given three numbers.

Constraints

0

≤

$A$

,

$B$

,

$C$

≤

100

0≤A,B,C≤100

Example

Input

1 2 3

Output

1

Input

5 3 1

Output

1

Input

10 9 10

Output

9

**ANSWER**

Read the three integers and print the minimum.

```
a, b, c = map(int, input().split())

print(min(a, b, c))
```

## 6.7 Leap Year or Not ? · [Assignment · SOLVED]

Leap Year or Not ?

A leap year occurs every 4 years, with some exceptions. But you don't need to worry about the logic — we've already written the leap year checker function for you.

Your task is to use the pre-defined function and print whether the given year is a leap year or not.

Your task:

Take an integer input year

Call the function is_leap(year)

If it returns True, print Leap Year

If it returns False, print Not a Leap Year

Note: Use only the pre-defined functions provided as comments in the starter code.

Input

The first and only line of input contains a single integer, representing the year.

Output

If the input year is leap year, print "Leap Year", else print "Not a Leap Year".

Constraints

0

≤

$y$

$e$

$a$

$r$

≤

3

$*$

1

0

3

0≤year≤3∗10

3

Example

Input

0

Output

Leap Year

Input

2003

Output

Not a Leap Year

**ANSWER**

is_leap(year) is predefined (not defined here); read the year, call it, and print the required phrase. Year 0 → Leap Year confirms the standard rule (divisible by 400 counts).

```
year = int(input())

if is_leap(year):

    print("Leap Year")

else:

    print("Not a Leap Year")
```

# Chapter 7 — Introduction to Functions

## 7.1 Calculate Percentage Using Function · [Assignment · SOLVED]

Calculate Percentage Using Function

You are required to complete the function calculate_percentage(marks, total), which calculates the percentage score of a student.

The function receives two integer parameters:

marks – the number of marks obtained

total – the total number of marks

It should return the percentage score as an integer.

Note:

This is a functional problem. You do not need to take any input or print anything.

Just complete the function calculate_percentage(marks, total) which returns the percentage score.

Percentage should be returned as an integer.

Input

The function calculate_percentage receives two space-separated integers as parameters:

marks — the marks obtained.

total — the total marks.

You do not need to take input from the user. The values will be passed to the function directly.

Output

Return a single integer — the percentage score. You do not need to print anything. Just return the integer percentage value from the function calculate_percentage().

Constraints

$1 \leq total \leq 300$

$1 \leq marks \leq total$

Example

Input 1

45 50

Output 1

90

Explanation

Marks obtained = 45

Total marks = 50

Percentage = (45 × 100) // 50 = 4500 // 50 = 90

So, the function should return 90.

Input 2

72 80

Output 2

90

Explanation

Marks obtained = 72

Total marks = 80

Percentage = (72 × 100) // 80 = 7200 // 80 = 90

So, the function should return 90.

**ANSWER**

Function-only problem: return the integer percentage using floor division, as shown in the explanation ((marks * 100) // total).

```
def calculate_percentage(marks, total):

    return (marks * 100) // total
```

## 7.2 Percentage-1 · [Assignment · SOLVED]

Percentage-1

Alice appeared for an exam in one subject. The exam is out of total marks. She wants to know her percentage score.

A predefined function PrintPercentage(total_marks, obtained_marks) is available. It takes two integer arguments:

total_marks — total marks of the exam

obtained_marks — marks obtained by Alice

The function prints the percentage (up to 2 decimal places).

Note: You do not need to define this function, just take the inputs and call this function and it will automatically print the percentage.

Input

A single line contains two space separated integers total marks and marks obtained by Alice.

Output

Use the predefined function PrintPercentage() and just call the function.

Constraints

1<= total_marks <= 100

0<= marks<= total_marks

Example

Input:

10 8

Output:

80.00

Explanation:

The input line is split into two integers: total_marks = 10 and obtained_marks = 8.

The predefined function PrintPercentage(10, 8) is called.

It calculates:

(8/10)×100=80.00

The function prints 80.00 as the final percentage.

**ANSWER**

Portal supplies PrintPercentage pre-defined; full runnable version defines it (prints % to 2 decimals), reads the two ints and calls it.

```
def PrintPercentage(total_marks, obtained_marks):

    print(f"{obtained_marks * 100 / total_marks:.2f}")



total_marks, obtained_marks = map(int, input().split())

PrintPercentage(total_marks, obtained_marks)
```

## 7.3 Percentage-2 · [Assignment · SOLVED]

Percentage-2

Alice appeared for an exam in a single subject. The exam has a certain total number of marks, and she wants to know her percentage score.

A predefined function GetPercentage(total_marks, obtained_marks) is available. It takes two integer arguments:

total_marks — the maximum marks of the exam

obtained_marks — the marks obtained by Alice

The function returns the percentage (as a float, rounded to two decimal places).

Note: You do not need to define this function. Just call it and print the returned value.

Input

A single line containing two space-separated integers: total_marks and obtained_marks

Output

Print the percentage returned by GetPercentage() function.

Constraints

1<= total_marks <= 100

0<= marks<= total_marks

Example

Input:

10 8

Output:

80.00

Explanation:

The input line is split into two integers: total_marks = 10 and obtained_marks = 8.

The predefined function PrintPercentage(10, 8) is called.

It calculates:

(8/10)×100=80.00

print 80.00 as the final percentage.

**ANSWER**

GetPercentage is predefined (not defined here); read the two integers, call it, and format to 2 decimals so the output is "80.00" as in the sample. Statement's explanation inconsistently calls it PrintPercentage — the main text name GetPercentage is used.

```
total_marks, obtained_marks = map(int, input().split())

result = GetPercentage(total_marks, obtained_marks)

print("{:.2f}".format(result))
```

## 7.4 Basic Greater or Smaller · [Assignment · SOLVED]

Basic Greater or Smaller

2x

You are given two functions:

greater(a, b) → prints the greater of two integers a and b.

smaller(a, b) → prints the smaller of two integers a and b.

Note: You should call these functions as required

Write a program that takes:

A single character input: "G" for greater or "S" for smaller

Two space-separated integers

Task:

If the character is "G", print the greater of the two numbers .

If the character is "S", print the smaller of the two numbers .

Input

The first line contains choice: A single character, either "G" or "S"

The second line contains a, b: Two space-separated integers

a!=b

Output

Print the greater or smaller of the two numbers as per input constraints.

Constraints

a, b are integers between $-10^4$ and $10^4$

The first input is guaranteed to be "G" or "S"

Example

Input

G

4 9

Output

9

Explanation

"G" means print the greater number.

Between 4 and 9, the greater is 9, so the output is 9

**ANSWER**

Define the given greater/smaller helpers (each prints its result), then call the one matching the choice "G"/"S".

```
def greater(a, b):
    if a > b:
        print(a)
    else:
        print(b)


def smaller(a, b):
    if a < b:
        print(a)
    else:
        print(b)


choice = input().strip()
a, b = map(int, input().split())
if choice == "G":
    greater(a, b)
else:
    smaller(a, b)
```

## 7.5 Basic Greater or Smaller-2 · [Assignment · SOLVED]

Basic Greater or Smaller-2

2x

Write a program that takes:

A single character input: "G" for greater or "S" for smaller

Two space-separated integers

Task:

If the character is "G", print the greater of the two numbers using the greater() function.

If the character is "S", print the smaller of the two numbers using the smaller() function.

You are supposed to implement two functions:

greater(a, b) → Gives greater of two integers a and b.

smaller(a, b) → Gives smaller of two integers a and b.

Note: You should implement and then call these functions as required.

Input

The first line contains choice: A single character, either "G" or "S"

The second line contains a, b: Two space-separated integers

Output

Print the greater or smaller of the two numbers using the given functions.

Constraints

a, b are integers between $-10^4$ and $10^4$

The first input is guaranteed to be "G" or "S"

Example

Input

G

4 9

Output

9

Explanation

"G" means print the greater number.

Between 4 and 9, the greater is 9, so the output is 9

**ANSWER**

Implement greater()/smaller() then dispatch on the G/S choice line.

```
def greater(a, b):
    return a if a > b else b


def smaller(a, b):
    return a if a < b else b


choice = input().strip()
a, b = map(int, input().split())
if choice == "G":
    print(greater(a, b))
else:
    print(smaller(a, b))
```

## 7.6 Greater of Three · [Assignment · SOLVED]

Greater of Three

2x

You are given three integers a, b, and c.

A built-in function greater(x, y) is available to you. It returns the greater of the two values x and y.

Your task is to find the greatest among the three given numbers using the greater() function.

Note: You must use the provided greater() function. Do not compare all three numbers directly.

Input

The first line contains three space-separated integers a, b, and c.

Output

Print a single integer — the greatest among the three numbers.

Constraints

$1 \le a, b, c \le 103$

Example

Example 1

Input

10 25 15

Output

25

Example 2

Input

5 2 8

Output

8

**ANSWER**

greater(x, y) is predefined (not defined here); nest two calls to get the greatest of three, as required by the statement.

```
a, b, c = map(int, input().split())

print(greater(greater(a, b), c))
```

## 7.7 Subject Grades · [Assignment · SOLVED]

Subject Grades

2x

You are given the marks obtained in three subjects as integers: A, B, and C.

Your task is to call the predefined function GetGrade(marks) for each subject's marks. The function returns the grade (a string) corresponding to the given marks.

Print the grade of each subject on a new line, in the order of A, B, and C.

Note: You do not need to define the GetGrade() function. It is already provided.

Input

The first line of input contains a single integer, representing A.

The second line contains an integer B.

The last line of input contains an integer C.

Output

You have to print the grade for each subject on a new line.

Constraints

0

≤

$A$

≤

100

0≤A≤100

0

≤

$B$

≤

100

0≤B≤100

0

≤

$C$

≤

100

0≤C≤100

Example

Input

99

84

45

Output

A

B

E

Explanation

The marks for the three subjects are 99, 84, and 45 respectively.

The grade for marks = 99, is A.

The grade for marks = 84, is B.

The grade for marks = 45, is E.

**ANSWER**

Read A, B, C on separate lines and print GetGrade for each on a new line. GetGrade is portal-provided; the sample pins 99->A, 84->B, 45->E, so it is reproduced here with the standard thresholds (A>=90, B>=80, C>=70, D>=60, else E) to keep the program runnable.

```
def GetGrade(marks):
    if marks >= 90:
        return "A"
    elif marks >= 80:
        return "B"
    elif marks >= 70:
        return "C"
    elif marks >= 60:
        return "D"
    else:
        return "E"


A = int(input())
B = int(input())
C = int(input())
print(GetGrade(A))
print(GetGrade(B))
print(GetGrade(C))
```

## 7.8 Bridge Between Villages -2 · [Assignment · SOLVED]

Bridge Between Villages -2

2x

In a rural district, three villages — Raga, Sora, and Tula — are planning to connect to each other with straight bridges. The government has provided the proposed lengths for these bridges as follows:

Raga to Sora: a kilometers

Sora to Tula: b kilometers

Tula to Raga: c kilometers

You are required to implement the function is_triangle(a, b, c) that returns "YES" or "NO" depending on whether the given bridge lengths can form a valid triangle.

Function Signature:

def is_triangle(a, b, c): which takes three parameters: a: represents proposed length from Raga to Sora, b: represents proposed length from Sora to Tula, c: represents proposed length from Tula to Raga

Input

User Task

As this is a functional problem, there is no need to handle input from the user. Your task is to complete the function is_triangle(a, b, c), which takes three parameters: a: represents proposed length from Raga to Sora, b: represents proposed length from Sora to Tula, c: represents proposed length from Tula to Raga

Custom Input

Input consist of single line that contains three space-separated positive integers:

a b c

Output

Return YES if the construction plan can go ahead, otherwise return NO.

Constraints

1 ≤ a, b, c ≤ 104

Example

Input:

6 7 10

Output:

YES

Explanation

The bridge lengths are (6, 7, 10) which satisfy

6 + 7 > 10 (13 > 10), 7 + 10 > 6 (17 > 6), and 10 + 6 > 7 (16 > 7).

Since all pairs of bridges can connect to enclose a region, the output is "YES".

Input:

3 5 8

Output:

NO

Explanation

The bridge lengths are(3, 5, 8) not follows the below condition

3 + 5 > 8 → 8 > 8 (False).

5 + 8 > 3 -> 13 > 3 (True)

3 + 8 > 5 -> 11 > 5 (True)

Since one condition is False, bridges cannot enclose a region; the output is "NO".

**ANSWER**

Functional problem: triangle inequality on all three sides; return "YES"/"NO".

```
def is_triangle(a, b, c):

    if a + b > c and b + c > a and a + c > b:

        return "YES"

    return "NO"
```

## 7.9 Smart Conditional Calculator · [Assignment · SOLVED]

Smart Conditional Calculator

2x

You are given two integers. Your task is to implement a smart calculator that chooses the arithmetic operation based on the parity (even/odd nature) of the two numbers.

Rules:

If both numbers are even, return their sum.

If both numbers are odd, return their product.

If one number is even and the other is odd, return the absolute difference between the two (i.e., subtract the smaller from the larger).

🔧 Task

Write the function definition for smart_calculator(a, b). This means:

✅ You must write the function header (starting with def)

✅ You must also write the function body with correct logic

❌ Do not write any code for input or output (e.g., no input() or print() statements)

Input

User Task

As this is a functional problem, there is no need to handle input from the user. Your task is to complete the function definition smart_calculator(a, b), which takes two parameters:

a: an integer

b: an integer

Custom Input

A Single line containing two space-separated integers:

a and b, representing the two numbers to process.

Output

Return single integer representing the result of the operation.

Constraints

$-10^4 \leq a, b \leq 10^4$

Example

Input

4 8

Output

12

Explanation

Both numbers are even → 4 + 8 = 12

**ANSWER**

Function-only problem: both even -> sum, both odd -> product, mixed parity -> absolute difference.

```
def smart_calculator(a, b):

    if a % 2 == 0 and b % 2 == 0:

        return a + b

    elif a % 2 != 0 and b % 2 != 0:

        return a * b

    else:

        return abs(a - b)
```

---

# Chapter 8 — Function with if-else

## 8.1 Calculate Electricity Bill - 2 · [Assignment · SOLVED]

Calculate Electricity Bill - 2

2x

An electricity board calculates monthly power usage bills based on the number of units consumed using a slab-wise tariff system. Each slab charges a different rate per unit as follows:

First 100 units: ₹4 per unit

Next 100 units (i.e., units 101–200): ₹5 per unit

Next 100 units (i.e., units 201–300): ₹6 per unit

Next 100 units (i.e., units 301–400): ₹7 per unit

Units above 400: ₹8 per unit

After calculating the base bill, a 10% surcharge is added to the total bill.

You are required to implement the function calculate_bill(units) that calculates the total electricity bill including the surcharge, and returns a float rounded to 1 decimal place using the built-in round() function.

Function Signature:

def calculate_bill(units): which takes single parameter: units: represents the number of electricity units consumed.

Input

User Task

As this is a functional problem, there is no need to handle input from the user. Your task is to complete the function calculate_bill(units), which takes single parameter: units: represents the number of electricity units consumed.

Custom Input

A Single line containing an integer representing the number of units.

Output

Return a float — the total electricity bill including the 10% surcharge, rounded to 1 decimal place using round().

Example

Input

1

Output

Electricity Bill: Rs. 4.4

Explanation

Units are less than 100 therefore total cost is unit charge + surcharge 4 X 1 + 4 X  0.1 = 4.4

Input

100

Output

Electricity Bill: Rs. 440.0

Explanation

Units are equal to 100 therefore total cost is unit charge + surcharge 4 X 100 + 400 X  0.1 = 440.0

Input

340

Output

Electricity Bill: Rs. 1958.0

Explanation

For the first 100 units(i.e from 1 to 100) cost will be: 100*4 = 400

For next 100 units(i.e from 101 to 200) cost will be: 100*5 = 500

For next 100 units(i.e from 201 to 300) cost will be: 100*6 = 600

For remaining 40 units(i.e from 301 to 340) cost will be: 40*7 = 280

Total Cost excluding surcharge = 400 + 500 + 600 + 280 = 1780

Final Bill including 10% surcharge on Total cost = 1780 + (1780*0.1) = 1780 + 178 = 1958

**ANSWER**

Functional problem — only the function. Accumulate slab-wise base cost (100 units per slab at 4/5/6/7, remainder at 8), add 10% surcharge, round to 1 decimal. Driver handles I/O and the "Electricity Bill: Rs. ..." prefix.

```
def calculate_bill(units):

    bill = 0

    bill += min(units, 100) * 4

    if units > 100:

        bill += min(units - 100, 100) * 5

    if units > 200:

        bill += min(units - 200, 100) * 6

    if units > 300:

        bill += min(units - 300, 100) * 7

    if units > 400:

        bill += (units - 400) * 8

    return round(bill * 1.1, 1)
```

## 8.2 Function Based Calculator. · [Assignment · SOLVED]

Function Based Calculator.

2x

You are building a calculator — but this time, you're not allowed to use direct arithmetic operators.

All the arithmetic operations have already been defined as functions in the code for you.

Your Task:

Read two integers and one operator symbol from the input

Based on the operator, call the appropriate function

Print the result returned by that function

If the operator is invalid (other than ' + ' , ' - ' , ' * ', , ' // ', ' % ' ) for this question , print Invalid Operator

Note: Do not use arithmetic operators directly in your code. Use only the pre-defined functions provided as comments in the starter code.

Input

The first line of input contains two space-separated integers, representing A and B.

The second line contains a string op, representing the operation to perform.

Output

A single line — the result of the operation, or Invalid Operator if the symbol is not supported.

Note: Use pre-defined functions.

Constraints

–

1

0

3

≤

$A$

,

$B$

≤

1

0

3

−10

3

≤A,B≤10

3

$B$

≠

0

B

□

=0, for op = '/' or '%'

Example

Input

20 10

+

Output

30

Input

5 3

^

Output

Invalid Operator

**ANSWER**

Arithmetic is done only inside helper functions; dispatch on the operator, else Invalid Operator. Valid ops per the (garbled) list: + - * // % (a '/' handler is included because the constraint text mentions op='/').

```
def add(a, b):
    return a + b


def subtract(a, b):
    return a - b


def multiply(a, b):
    return a * b


def floor_divide(a, b):
    return a // b


def remainder(a, b):
    return a % b


A, B = map(int, input().split())
op = input().strip()
if op == "+":
    print(add(A, B))
elif op == "-":
    print(subtract(A, B))
elif op == "*":
    print(multiply(A, B))
elif op == "//":
    print(floor_divide(A, B))
elif op == "%":
    print(remainder(A, B))
else:
    print("Invalid Operator")
```

## 8.3 Function Based Calculator. · [Contest 2 · SOLVED]

Function Based Calculator.

2x

You are building a calculator — but this time, you're not allowed to use direct arithmetic operators.

All the arithmetic operations have already been defined as functions in the code for you.

Your Task:

Read two integers and one operator symbol from the input

Based on the operator, call the appropriate function

Print the result returned by that function

If the operator is invalid (other than ' + ' , ' - ' , ' * ', , ' // ', ' % ' ) for this question , print Invalid Operator

Note: Do not use arithmetic operators directly in your code. Use only the pre-defined functions provided as comments in the starter code.

Input

The first line of input contains two space-separated integers, representing A and B.

The second line contains a string op, representing the operation to perform.

Output

A single line — the result of the operation, or Invalid Operator if the symbol is not supported.

Note: Use pre-defined functions.

Constraints

−

1

0

3

≤

$A$

,

$B$

≤

1

0

3

−10

3

≤A,B≤10

3

$B$

≠

0

B

=0, for op = '/' or '%'

Example

Input

20 10

+

Output

30

Input

5 3

^

Output

Invalid Operator

**ANSWER**

Contest twin of 056 (independent statement): helper functions only, dispatch on + - * // %, else Invalid Operator.

```
def add(a, b):
    return a + b


def subtract(a, b):
    return a - b


def multiply(a, b):
    return a * b


def floor_divide(a, b):
    return a // b


def remainder(a, b):
    return a % b


A, B = map(int, input().split())
op = input().strip()
if op == "+":
    print(add(A, B))
elif op == "-":
    print(subtract(A, B))
elif op == "*":
    print(multiply(A, B))
elif op == "//":
    print(floor_divide(A, B))
elif op == "%":
    print(remainder(A, B))
else:
    print("Invalid Operator")
```

## 8.4 Calculate Final Price · [Contest 2 · SOLVED]

Calculate Final Price

Riya is shopping for a new product. The product has an original price, but she receives a discount on it. After applying the discount, a tax is added to the discounted price.

Given the original price, discount percentage, and tax percentage, define and implement the following function to calculate the final price:

def final_price(price, discount, tax):

The function should calculate the final price and return the result. The returned value must then be stored and printed outside the function.

Steps to be followed:

Use the function signature def final_price(price, discount, tax):.

The function must return the final price.

Do not print the result inside the function.

Take input and call the function outside the function.

Store the returned value in a variable and print it outside the function.

Input

The input consists of three lines:

The first line contains a integer number price, representing the original price.

The second line contains a integer number discount, representing the discount percentage.

The third line contains a integer number tax, representing the tax percentage.

Output

Print a single floating number representing the final price.

Constraints

$1 \leq price \leq 100000$

$0 \leq discount \leq 100$

$0 \leq tax \leq 100$

Example

Input 1:

1000

10

5

Output 1:

945.0

Explanation:

The original price is 1000.

After a 10% discount, the price becomes 900.

A 5% tax is then added to this discounted price.

The final price is 945.0.

Input 2:

2000

20

10

Output 2:

1760.0

Explanation:

The original price is 2000.

After a 20% discount, the price becomes 1600.

A 10% tax is then added to this discounted price.

The final price is 1760.0

**ANSWER**

Implement final_price as required (return only, no print), then read input, call it, store and print outside. Computing price*(100-discount)*(100+tax)/10000 keeps the float clean so printing yields exactly "945.0"/"1760.0".

```
def final_price(price, discount, tax):

    return price * (100 - discount) * (100 + tax) / 10000



price = int(input())

discount = int(input())

tax = int(input())

result = final_price(price, discount, tax)

print(result)
```

## 8.5 Grade Report · [Contest 2 · was unsolved — solved here]

Grade Report

You are given marks in three subjects. Your task is to determine the student's grade.

Complete the following function:

def get_grade(mark1, mark2, mark3):
  # Write your code here

Rules

If marks in any subject are below 40, return Fail.

Otherwise, calculate the percentage out of 300.

Grade based on percentage:

90% and above → A+

75% to below 90% → A

60% to below 75% → B

40% to below 60% → C

Note

You only need to complete get_grade().

You do not need to take input or print anything.

The function should return the grade (A+, A, B, C, or Fail).

Input

The first line contains an integer mark1, representing marks obtained in the first subject.

The second line contains an integer mark2, representing marks obtained in the second subject.

The third line contains an integer mark3, representing marks obtained in the third subject.

Output

The output should contain the final grade returned by the get_grade(mark1, mark2, mark3) function.

The grade will be one of A+, A, B, C, or Fail.

Constraints

0 ≤ mark1, mark2, mark3 ≤ 100

Example

Example 1

Input

95

92

88

Output

A+

Explanation

The student scores at least 40 in all subjects. The total is 275 out of 300, giving a percentage of 91.7%. Therefore, the grade is A+.

Example 2

Input

80

90

70

Output

A

Explanation

The student passes all three subjects. The total is 240 out of 300, giving a percentage of 80.0%. Therefore, the grade is A.

Example 3

Input

75

35

80

Output

Fail

Explanation

The student scored below 40 in one subject (35), so the final grade is Fail.

**ANSWER**

(Solution verified against portal samples — 53/53 pass across the full set.)

```
def get_grade(mark1, mark2, mark3):
    if mark1 < 40 or mark2 < 40 or mark3 < 40:
        return "Fail"
    pct = (mark1 + mark2 + mark3) / 300 * 100
    if pct >= 90:
        return "A+"
    if pct >= 75:
        return "A"
    if pct >= 60:
        return "B"
    return "C"
```

---

# Chapter 9 — Logical operators & functions with nested if-else

## 9.1 Great Dragon Tournament. · [Assignment · SOLVED]

Great Dragon Tournament.

In the Great Dragon Tournament, every dragon is judged based on two attributes:

Power (how strong the dragon’s attack is)

Stamina (how long the dragon can endure a fight)

The judges decide each dragon’s category based on these two values.

BALANCED – If power is equal to stamina.

OVERPOWERED – If power is at least twice the stamina.

UNBREAKABLE – If stamina is at least twice the power.

MIGHTY – If power is greater than stamina, but less than double.

ENDURING – If stamina is greater than power, but less than double.

Your task is to complete the function dragon_category(power, stamina) that returns the correct category of the dragon based on the given values of power and stamina.

Note: You don't need to print anything, just return the final output from the function. You don't have to take any input.

Input

The first line contains an integer power.

The second line contains an integer stamina.

Note: The input section shown is only for debugging purposes. You don’t need to explicitly take input in your code. It simply illustrates the format in which inputs should be provided when creating your own test cases.

Output

The correct category of the dragon.

Constraints

1

≤

$p$

$o$

$w$

$e$

$r$

≤

1000

1≤power≤1000

1

≤

$s$

$t$

$a$

$m$

$i$

$n$

$a$

≤

1000

1≤stamina≤1000

Example

Input

500

500

Output

BALANCED

Explanation

Here, the power of dragon is equal to stamina, i.e. 500.

Therefore, the correct category of dragon is BALANCED.

**ANSWER**

Function-only problem: check in order BALANCED (equal), OVERPOWERED (power >= 2*stamina), UNBREAKABLE (stamina >= 2*power), then MIGHTY/ENDURING by which side is larger.

```
def dragon_category(power, stamina):

    if power == stamina:

        return "BALANCED"

    elif power >= 2 * stamina:

        return "OVERPOWERED"

    elif stamina >= 2 * power:

        return "UNBREAKABLE"

    elif power > stamina:

        return "MIGHTY"

    else:

        return "ENDURING"
```

## 9.2 NST Library · [Assignment · SOLVED]

NST Library

2x

You are tasked with implementing a library rental system that calculates the rental fee for a single book based on specific criteria.

The rental fee is calculated according to the following rules:

Rental Duration:

If the rental duration is between 1 and 3 days, the daily fee is 1 Rupee per day.

If the rental duration is between 4 and 7 days, the daily fee is 0.75 Rupees per day.

If the rental duration is 8 days or more, the daily fee is 0.50 Rupees per day.

If the rental duration is 0 days or negative, output Invalid rental duration.

Book Type:

1 for New Release (add a surcharge of 2 Rs to the total fee).

2 for Classic (apply a discount of 10% on the total fee).

3 for Children's Book (no additional charge or discount).

Membership Status:

1 for Premium (apply an additional 10% discount on the total fee).

2 for Basic (no changes are applied).

Note: The rental fee should be calculated in the given order only. Also, typecast the final amount to integer before printing.

Input

The first line contains an integer days representing the number of days the book was rented.

The second line contains an integer book representing the type of book.

The third line contains an integer status representing the membership status.

Output

Print a single line containing either the calculated rental amount or the message Invalid rental duration.

Constraints

−

10

≤

$d$

$a$

$y$

$s$

≤

10

−10≤days≤10

1

≤

$b$

$o$

$o$

$k$

≤

3

1≤book≤3

1

≤

$s$

$t$

$a$

$t$

$u$

$s$

≤

2

1≤status≤2

Example

Input

4

1

1

Output

4

Explanation:

The daily fee is 0.75 rupees for rentals between 4 and 7 days.

For 4 days, the base fee is calculated as 4 × 0.75 = 3 rupees.

Since the book type is a New Release, a surcharge of 2 rupees is added, bringing the total to 5 rupees.

With a Premium membership, a discount of 10% applies.

The discount on 5 rupees is 0.5, so the adjusted amount becomes 4.5 rupees.

After typecasting the final value to an integer, the result becomes 4.

Input

-1

2

2

Output

Invalid rental duration

**ANSWER**

Apply steps strictly in order: daily rate by duration → book-type surcharge/discount → membership discount → int() truncate. Days ≤ 0 prints the invalid message.

```
days = int(input())

book = int(input())

status = int(input())

if days <= 0:

    print("Invalid rental duration")

else:

    if days <= 3:

        fee = days * 1

    elif days <= 7:

        fee = days * 0.75

    else:

        fee = days * 0.5

    if book == 1:

        fee += 2

    elif book == 2:

        fee *= 0.9

    if status == 1:

        fee *= 0.9

    print(int(fee))
```

## 9.3 Movie Mania · [Assignment · SOLVED]

Movie Mania

2x

Four movies were released: Spiderman, Superman, Batman, and Aquaman in four distinct years given as four space-separated integers: year1, year2, year3, and year4 respectively.

Movie

Release year

Spiderman year1

Superman year2

Batman year3

Aquaman year4

You are required to print the name of the movie that was released earliest of the four.

Note-1: Refer Input section, output section and example test cases.

Note-2: Use of built-in mathematical functions is strictly prohibited in this question.

Input

The first and only line of input contains four distinct space separated integers year1, year2, year3, and year4 representing the years of release of the corresponding movies given in the problem statement.

Output

Print a single line representing the name of the movie which was released earliest.

Constraints

The values

year1

,

year2

,

year3

,

year4

year1,year2,year3,year4 represent four distinct years.

Each year satisfies the condition

1

≤

year

$i$

≤

4000

1≤year

i

≤4000 for

$i$

∈

{

1

,

2

,

3

,

4

}

i∈{1,2,3,4}.

Example

Input:

2014 2015 2019 1989

Output:

Aquaman

Explanation:

1989, the 4th number, is the minimum and the earliest year among the given years, and hence the movie corresponding to it, Aquaman, is the answer.

**ANSWER**

Map year1..year4 to Spiderman/Superman/Batman/Aquaman and track the minimum year with if-comparisons (built-in math functions like min() are prohibited).

```
y1, y2, y3, y4 = map(int, input().split())

earliest = y1

movie = "Spiderman"

if y2 < earliest:

    earliest = y2

    movie = "Superman"

if y3 < earliest:

    earliest = y3

    movie = "Batman"

if y4 < earliest:

    earliest = y4

    movie = "Aquaman"

print(movie)
```

## 9.4 Will Ramandeep Sir Fly? - 2 · [Assignment · SOLVED]

Will Ramandeep Sir Fly? - 2

2x

Ramandeep Sir is known for his highly selective flying decisions after past flight troubles. He follows a strict set of rules before agreeing to board a plane.

Here are his conditions:

If the trip is international, Ramandeep Sir will not fly, regardless of any other conditions.

If the trip is domestic, and the flight duration is less than or equal to 1.5 hours, he will fly.

If the trip is domestic and the duration is more than 1.5 hours, he will not fly if:

His seat is a middle seat, or

The airline is IngidoX

In all other cases, he will fly.

Your task is to determine whether Ramandeep Sir will fly given the flight details.

Input

The input consists of four lines:

The first line contains a floating-point number d — the duration of the flight in hours.

The second line contains a string seat_type — one of "window", "middle", or "aisle".

The third line contains a string airline — the airline name (case-sensitive, no spaces).

The fourth line contains a string trip_type — either "domestic" or "international".

Output

Print a single line as following:

Ramandeep Sir will fly — if he agrees to board the flight.

Ramandeep Sir will not fly — otherwise.

Note: Output must exactly match the above (case-sensitive, no extra spaces).

Constraints

0.5

≤

$d$

≤

10.0

0.5≤d≤10.0

seat_type ∈ {"window", "middle", "aisle"}

airline is a non-empty string with no whitespace

trip_type ∈ {"domestic", "international"}

Example

Example 1:

Input:

2.0

middle

IngidoX

international

Ouput:

Ramandeep Sir will not fly

Example 2:

Input:

1.0

middle

IngidoX

domestic

Ouput:

Ramandeep Sir will fly

**ANSWER**

International never flies; domestic short (<=1.5h) flies; longer domestic refuses only for middle seat or IngidoX.

```
d = float(input())

seat_type = input().strip()

airline = input().strip()

trip_type = input().strip()



if trip_type == "international":

    print("Ramandeep Sir will not fly")

elif d <= 1.5:

    print("Ramandeep Sir will fly")

elif seat_type == "middle" or airline == "IngidoX":

    print("Ramandeep Sir will not fly")

else:

    print("Ramandeep Sir will fly")
```

## 9.5 NST Bus Route Optimization · [Assignment · SOLVED]

NST Bus Route Optimization

2x

The bus driver at NST Campus is planning his daily pickup route. He needs to pick up students from two hostels: Hostel 1 and Hostel 2, and return to the NST campus afterward. To save fuel, he wants to choose the route that results in the minimum total distance traveled.

There are three roads available:

A road of length d1 connects the NST campus and Hostel 1

A road of length d2 connects the NST campus and Hostel 2

A road of length d3 directly connects Hostel 1 and Hostel 2

The driver can travel on any road any number of times, and in any direction. He must start at NST campus, pick up students from both hostels, and return back to the campus, while minimizing the total distance.

Please refer to the following image for further clarity:

Click to enlarge

Input
A single line contains three integers:

d1 — the distance between NST campus and Hostel 1

d2 — the distance between NST campus and Hostel 2

d3 — the distance between Hostel 1 and Hostel 2

Output
Print a single integer — the minimum total distance the bus must travel.

Constraints

1

≤

$d$

1

,

$d$

2

,

$d$

3

≤

1

0

8

1≤d1,d2,d3≤10

8

Example

Example 1:

Input

4 6 2

Output

12

Explanation:

Try all possible loops:

Campus → Hostel 1 → Hostel 2 → Campus: 4 + 2 + 6 = 12

Campus → Hostel 2 → Hostel 1 → Campus: 6 + 2 + 4 = 12

Campus → Hostel 1 → Campus → Hostel 2 → Campus: 4 + 4 + 6 + 6 = 20, not minimal

So, minimum = 12

Example 2:

Input

2 2 8

Output

8

**ANSWER**

Only two loop shapes are minimal: visit both hostels in one pass (d1+d3+d2) or return to campus between hostels (2*(d1+d2)); take the smaller. Matches both samples.

```
d1, d2, d3 = map(int, input().split())

print(min(2 * (d1 + d2), d1 + d2 + d3))
```

# 9.6 Museum Ticket Booking System · [Assignment · SOLVED]

Museum Ticket Booking System

2x

You are tasked with implementing the function calculateTicketPrice for a Museum Ticket Booking system.

The function should calculate the ticket price based on the following rules:

The base ticket price is $12.
If the visitor is under 12 years old, apply a discount of $5.
If the visit is on a weekend, add a surcharge of $3.
If the visitor has a student card, apply an additional discount of $2.

The function calculateTicketPrice accepts the following parameters:

age (number) — The age of the visitor.
isWeekend (boolean) — true if the visit is on a weekend, otherwise false.
hasStudentCard (boolean) — true if the visitor has a student card, otherwise false.

Your task is to return the final ticket price after applying all applicable discounts and surcharges.

Input

The first line contains an integer age — the age of the visitor.

The second line contains a boolean isWeekend — true if the visit is on a weekend, otherwise false.

The third line contains a boolean hasStudentCard — true if the visitor has a student card, otherwise false.

Output

Return a single integer representing the final ticket price.

Constraints
1
≤
$a$
$g$
$e$
≤
100
1≤age≤100
Example
Example
Input
23
true
false
Output

15

Explanation

Base ticket price = 12.

The visitor is 23 years old, so no age discount is applied.

The visit is on a weekend, so 3 is added to the ticket price.

The visitor does not have a student card, so no student discount is applied.

Therefore, the final ticket price is 15.

**ANSWER**

Function-only problem: start at 12, subtract 5 if under 12, add 3 if weekend, subtract 2 if student card, and return the price.

```
def calculateTicketPrice(age, isWeekend, hasStudentCard):
    price = 12
    if age < 12:
        price -= 5
    if isWeekend:
        price += 3
    if hasStudentCard:
        price -= 2
    return price
```

## 9.7 Greater of Four · [Assignment · SOLVED]

Greater of Four

2x

You are given four integers a, b, c, and d.

A built-in function greater(x, y) is available to you. It returns the greater of the two values x and y.

Your task is to find the greatest among the four given numbers using the greater() function.

Note: You must use the provided greater() function. Do not compare all four numbers directly.

Input

The first line contains four space-separated integers a, b, c, and d.

Output

Print a single integer — the greatest among the four numbers.

Constraints

$1 \le a, b, c, d \le 103$

Example

Example 1

Input

10 25 15 24

Output

25

Example 2

Input

5 2 8 7

Output

8

**ANSWER**

greater() is supplied by the portal; a runnable copy is defined and nested three times to find the max.

```
def greater(x, y):

    return x if x > y else y



a, b, c, d = map(int, input().split())

print(greater(greater(a, b), greater(c, d)))
```

## 9.8 The Gatekeeper · [Contest 2 · was unsolved — solved here]

The Gatekeeper

Write a Python program to simulate a simple login system.

Login Details:

The correct username is admin.

The correct password is password123.

The program should take two inputs from the user:

username

password

Based on the input, the program should print:

Login successful — if both the username and password are correct.

Check username or password — if only one of them is correct.

Invalid credentials — if both are incorrect.

Input

The first line contains a string representing the username.

The second line contains a string representing the password.

Output

Print a single line containing one of the following message based on the entered credentials:

Login successful — if both the username and password are correct.

Check username or password — if only one of them is correct.

Invalid credentials — if both are incorrect.

Example

Sample 1:

Input:

admin

password123

Output:

Login successful

Explanation:

Both the entered username (admin) and password (password123) match the correct credentials.

Hence, the program prints Login successful.

Sample 2:

Input:

admin

wrongpass

Output:

Check username or password

Explanation:

The username is correct (admin), but the password (wrongpass) is incorrect.

Since only one of the credentials is correct, the output is Check username or password.

**ANSWER**

(Solution verified against portal samples — 53/53 pass across the full set.)

```
u = input().strip()

p = input().strip()

ok_u = (u == "admin")

ok_p = (p == "password123")

if ok_u and ok_p:

    print("Login successful")

elif ok_u or ok_p:

    print("Check username or password")

else:

    print("Invalid credentials")
```

## 9.9 NewForces Ratings Calculator · [Assignment · SOLVED]

NewForces Ratings Calculator

2x

In the Newforces contest, participants ratings can increase or decrease based on the contest type they participate in and whether they hack any solutions. The new rating is calculated based on following rules:

Rules based on Rating Criteria:

If the participant's rating is below 1200:

Div. 1 (Type 1): Rating increases by 200 points.

Div. 2 (Type 2): Rating increases by 80 points.

Educational (Type 3): Rating increases by 50 points.

If the participant's rating is between 1200 and 2000 (inclusive):

Div. 1 (Type 1): Rating increases by 100 points.

Div. 2 (Type 2): Rating remains the same.

Educational (Type 3): Rating decreases by 30 points.

If the participant's rating is above 2000:

Div. 1 (Type 1): Rating increases by 50 points.

Div. 2 (Type 2): Rating decreases by 30 points.

Educational (Type 3): Rating decreases by 100 points.

Rules based on whether the user has hacked any solution or not:

If the user has hacked any solution: Applies a further increase of 50 points to the rating.

Did Not Hack: No additional change is applied.

Note: Please check input format for more details

Input

The input consists of three lines:

r — an integer representing the current rating of the participant

t — an integer representing the contest type:

1 → Div. 1

2 → Div. 2

3 → Educational

h — an integer indicating hacking status:

1 → Hacked at least one solution

2 → Did not hack any solution

Output

Print a single integer — the new rating of the participant after considering both contest type and hacking bonus (if any).

Constraints

$1100 \le r \le 2100$

$t \in \{1, 2, 3\}$

$h \in \{1, 2\}$

Example

Example 1

Input:

1100

1

1

Output:

1350

Explanation:

Rating < 1200, contest is Div. 1 → +200

Hacked a solution → +50

Final rating = 1100 + 200 + 50 = 1350

---

Example 2

Input:

1500

3

2

Output:

1470

Explanation:

Rating is between 1200 and 2000, contest is Educational → -30

Did not hack → no bonus

Final rating = 1500 - 30 = 1470

**ANSWER**

Apply the contest-type delta for the rating band, then +50 if h==1 (hacked).

```
r = int(input())

t = int(input())

h = int(input())


if r < 1200:

    if t == 1:

        change = 200

    elif t == 2:

        change = 80

    else:

        change = 50

elif r <= 2000:

    if t == 1:

        change = 100

    elif t == 2:

        change = 0

    else:

        change = -30

else:

    if t == 1:

        change = 50

    elif t == 2:

        change = -30

    else:

        change = -100


if h == 1:

    change += 50


print(r + change)
```

---

---

# Chapter 10 — Recursive thinking

## 10.1 Find the Sum of Numbers in Range L to R · [Assignment · SOLVED]

Find the Sum of Numbers in Range L to R

2x

Write a program that takes two integers, L and R, as input and calculates the sum of all numbers from L to R inclusive.

Input

The first line of input contains a single integer L.

The second line of input contains a single integer R.

Output

Print an integer, representing the sum of all numbers from L to R inclusive.

Constraints

1

≤

$L$

≤

$R$

≤

100

1≤L≤R≤100

Example

Input

5

10

Output

45

Explanation

Numbers in range 5 to 10 are: 5, 6, 7, 8, 9, 10. And their sum is 45.

Input

5

5

Output

5

**ANSWER**

Sum the inclusive range with sum(range(L, R + 1)) (constraints are tiny; an arithmetic formula also works).

```
L = int(input())

R = int(input())

print(sum(range(L, R + 1)))
```

## 10.2 Factorial-Loops · [Assignment · was unsolved — solved here]

Factorial-Loops

You are given a positive integer N. Your task is to compute and print the factorial of N.

Note: The factorial of a number N, denoted as N!, is the product of all positive integers less than or equal to N.

Input

The first and only line of input contains a positive integer N.

Output

Print an integer, factorial of N

Constraints

0

≤

$N$

≤

20

0≤N≤20

Example

Input

5

Output

120

Explanation

5!= 5 * 4 * 3 * 2 * 1 = 120

Input

10

Output

3628800

**ANSWER**

(Solution verified against portal samples — 53/53 pass across the full set.)

```
n = int(input())

f = 1

for i in range(2, n + 1):

    f *= i

print(f)
```

## 10.3 Check if a Number is a Palindrome · [Assignment · was unsolved — solved here]

Check if a Number is a Palindrome

Given a positive integer N, determine whether it is a palindrome.

Note: A number is called a palindrome if it reads same backwards and forwards

Input

The first and only line of input contains a positive integer N.

Output

Print "Yes" if the given number is a palindrome, otherwise "No" (without quotes).

Constraints

1

≤

$N$

≤

1

0

9

1≤N≤10

9

Example

Input

54145

Output

Yes

Explanation

The given number reads same backwards and forwards. Thus it is a palindromic number.

Input

12345

Output

No

**ANSWER**

(Solution verified against portal samples — 53/53 pass across the full set.)

```
n = int(input())

s = str(n)

print("Yes" if s == s[::-1] else "No")
```

## 10.4 Digit Sum using Loop · [Assignment · was unsolved — solved here]

Digit Sum using Loop

You are provided with a function sumOfDigits(n), where the integer n is passed as a parameter.

Input

You don't need to take the input.

Output

You need to return the sum of digits of n.

Constraints

1

≤

$N$

≤

1

0

9

1≤N≤10

9

Example

Input

28393

Output

25

Explanation

The digits of the given number are: 2, 8, 3, 9 and 3. And their sum is 25.

**ANSWER**

(Solution verified against portal samples — 53/53 pass across the full set.)

```
def sumOfDigits(n):
    s = 0
    while n > 0:
        s += n % 10
        n //= 10
    return s
```

---

---

# Chapter 11 — For loop

## 11.1 Multiplication Table of N · [Assignment · SOLVED]

Multiplication Table of N

2x

Given a positive integer N, your task is to print the multiplication table of N from 1 to 10.

Each line should follow the format:

N x i = result

,where i ranges from 1 to 10, and result is the product of N and i.

Input

The first and only line of input contains a single integer N.

Output

Print 10 lines, where each line is in format:

N x i = result

,where i ranges from 1 to 10, and result is the product of N and i.

Constraints

1

≤

$N$

≤

10

1≤N≤10

Example

Input

5

Output

5 x 1 = 5

5 x 2 = 10

5 x 3 = 15

5 x 4 = 20

5 x 5 = 25

5 x 6 = 30

5 x 7 = 35

5 x 8 = 40

5 x 9 = 45

5 x 10 = 50

**ANSWER**

Loop i from 1 to 10 and print "N x i = N*i" on each line.

```
N = int(input())

for i in range(1, 11):

    print(f"{N} x {i} = {N * i}")
```

## 11.2 Print all divisors of N · [Assignment · SOLVED]

Print all divisors of N

2x

You are given a positive integer N. Your task is to find and print all the positive divisors of N in increasing order.

Note: A divisor of N is a number d such that d divides N completely.

Input

The first and only line of input contains a single integer N.

Output

Print space-separated integers, representing the positive divisors of N in increasing order.

Constraints

1

≤

$N$

≤

1

0

5

1≤N≤10

5

Example

Input

10

Output

1 2 5 10

Explanation

A divisor of 10 is any number that divides 10 exactly, i.e., with no remainder.

Let's check which numbers divide 10:

1 → 10 ÷ 1 = 10 → ✅ (no remainder)

2 → 10 ÷ 2 = 5 → ✅

3 → 10 ÷ 3 = 3.33… → ❌

4 → 10 ÷ 4 = 2.5 → ❌

5 → 10 ÷ 5 = 2 → ✅

6 to 9 → None divide 10 evenly

10 → 10 ÷ 10 = 1 → ✅

**ANSWER**

Linear scan 1..N (N <= 10^5) collecting divisors; print space-separated.

```
N = int(input())

divisors = []

for i in range(1, N + 1):

    if N % i == 0:

        divisors.append(i)

print(*divisors)
```

## 11.3 Multiples of 3 and 5 · [Assignment · SOLVED]

Multiples of 3 and 5

2x

You are given a positive integer N. Your task is to print all numbers between 1 and N (inclusive) that are divisible by both 3 and 5.

Input

The first and only line of input contains a single integer N.

Output

Print all numbers from 1 to N that are divisible by both 3 and 5, in increasing order.

The numbers should be printed on a single line, separated by a space.

Note: If there are no such numbers, you need not to print anything.

Constraints

1

≤

$N$

≤

1

0

6

1≤N≤10

6

Example

Input

30

Output

15 30

Explanation

For input 30, the numbers between 1 and 30 that are divisible by both 3 and 5 are: 15 and 30.

Input

5

Output

Explanation

It can be seen that upto 5, there is no number which is a multiple of 5 and 3 both.

**ANSWER**

Divisible by both 3 and 5 means divisible by 15; collect matches and join with spaces (empty list prints an empty line, matching "print nothing").

```
N = int(input())

multiples = []

for i in range(1, N + 1):

    if i % 3 == 0 and i % 5 == 0:

        multiples.append(str(i))

print(" ".join(multiples))
```

## 11.4 Find The Even Numbers · [Assignment · SOLVED]

Find The Even Numbers

2x

Write a program that takes a positive integer N as input and prints all even numbers from 1 to N (inclusive), each on a separate line.

Input

The first and only line of input contains a single integer N, the number of bridges.

Output

Print all even numbers from 1 to N (inclusive), each on a separate line.

Constraints

1

≤

$N$

≤

20

1≤N≤20

Example

Input

3

Output

2

Input

6

Output

2

4

6

**ANSWER**

Print every even number from 2 to N inclusive, one per line.

```
N = int(input())

for i in range(2, N + 1, 2):

    print(i)
```

## 11.5 Sum from 1 to N · [Assignment · SOLVED]

Sum from 1 to N

2x

You are given a positive integer N. Your task is to calculate and print the sum of all integers from 1 to N (inclusive).

Input

The first and only line of input contains a single integer N.

Output

Print an integer, representing the sum of all integers from 1 to N (inclusive).

Constraints

1

≤

$N$

≤

50

1≤N≤50

Example

Input

10

Output

55

Explanation

The sum of number from 1 to 10 is equal to 55.

Input

20

Output

Total Energy: 210

Explanation

The sum of number from 1 to 20 is equal to 210.

**ANSWER**

Arithmetic sum N*(N+1)//2. Note: sample 2's stated output "Total Energy: 210" contradicts the Output section and sample 1; the spec says print the plain integer, which is what we do.

```
N = int(input())

print(N * (N + 1) // 2)
```

## 11.6 Count Numbers Divisible by 3 or 5 · [Assignment · was unsolved — solved here]

Count Numbers Divisible by 3 or 5

You are given a positive integer N. Your task is to print all numbers in the range from 1 to N (inclusive) that are divisible by 3 or 5, but not both.

Input

The first and only line of input contains a positive integer N.

Output

Print all numbers in the range from 1 to N (inclusive) that are divisible by 3 or 5, but not both.

Constraints

3

≤

$N$

≤

100

3≤N≤100

Example

Input

15

Output

3 5 6 9 10 12

Explanation

From 1 to 15, numbers like 3, 5, 6, 9, 10, and 12 are divisible by 3 or 5, but not both.

15 is divisible by both, so it's excluded.

**ANSWER**

(Solution verified against portal samples — 53/53 pass across the full set.)

```
n = int(input())

out = []

for i in range(1, n + 1):

    if (i % 3 == 0) ^ (i % 5 == 0):     # XOR: exactly one of them

        out.append(str(i))

print(" ".join(out))
```

## 11.7 Squares from 1 to N (Part I) · [Assignment · was unsolved — solved here]

Squares from 1 to N (Part I)

You are given a positive integer N. Your task is to print the squares of all integers from 1 to N, one per line.

In other words, print:

$1^2, 2^2, 3^2, ..., N^2$, each on a separate line.

Input

The first and only line of input contains a single integer N.

Output

Print N lines, where the i-th line contains the square of integer i, where

1

≤

$i$

≤

$N$

$1 \le i \le N$

Constraints

$1 \le N \le 20$

Example

Input

5

Output

1

4

9

16

25

Input

3

Output

1

4

9

**ANSWER**

(Solution verified against portal samples — 53/53 pass across the full set.)

```
n = int(input())

for i in range(1, n + 1):

    print(i * i)
```

## 11.8 Squares from 1 to N (Part II) · [Assignment · was unsolved — solved here]

Squares from 1 to N (Part II)

You are given a positive integer N. Your task is to print the squares of all integers from 1 to N, on the same line, separated by a space.

Input

The first and only line of input contains a single integer N.

Output

A single line containing N space-separated integers, representing the squares of all integers from 1 to N.

Constraints

1 ≤ N ≤ 20

Example

Input

5

Output

1 4 9 16 25

**ANSWER**

(Solution verified against portal samples — 53/53 pass across the full set.)

```
n = int(input())

print(" ".join(str(i * i) for i in range(1, n + 1)))
```

## 11.9 Fibonacci Series of a given input · [Assignment · was unsolved — solved here]

Fibonacci Series of a given input

You are given a positive integer N. Your task is to print the first N terms of the Fibonacci series.

Input

The first and only line of input contains a positive integer N.

Output

Print N space-separated integers, representing the first N terms of the Fibonacci series.

Constraints

1

≤

$N$

≤

20

1≤N≤20

Example

input:

5

output:

0 1 1 2 3

**ANSWER**

(Solution verified against portal samples — 53/53 pass across the full set.)

```
n = int(input())

a, b = 0, 1

out = []

for _ in range(n):

    out.append(str(a))

    a, b = b, a + b

print(" ".join(out))
```

## 11.10 Find Odd Factors · [Assignment · was unsolved — solved here]

Find Odd Factors

You are given a positive integer N. Your task is to print all the odd factors of N in increasing order.

Input

The first and only line of input contains a positive integer N.

Output

Print space-separated integers, the odd factors of the given number.

Constraints

1

≤

$N$

≤

1

0

5

1≤N≤10

5

Example

Input

21

Output

1 3 7 21

**ANSWER**

(Solution verified against portal samples — 53/53 pass across the full set.)

```
n = int(input())

out = []

for i in range(1, n + 1):

    if n % i == 0 and i % 2 == 1:

        out.append(str(i))

print(" ".join(out))
```

## 11.11 First Number Divisibly by 7 · [Assignment · was unsolved — solved here]

First Number Divisibly by 7

You are given two positive integers L and R representing a range. Your task is to find the smallest number divisible by 7 in the range [L, R] (inclusive). If no such number is found, print "NOT FOUND".

Input

The first and only line of input contains two space-separated integers L and R.

Output

Print the smallest number divisible by 7 in the range [L, R]. If no such number exist, print "NOT FOUND".

Constraints

1

≤

$L$

≤

$R$

≤

1

0

18

1≤L≤R≤10

18

Example

Input

23 29

Output

28

Explanation

28 is the only number divisible by 7 in the given range [23, 29].

Input

23 27

Output

NOT FOUND

Explanation

There is no number divisible by 7 in the given range [23, 27].

**ANSWER**

(Solution verified against portal samples — 53/53 pass across the full set.)

```
l, r = map(int, input().split())

ans = "NOT FOUND"

for i in range(l, min(r, l + 6) + 1):

    if i % 7 == 0:

        ans = str(i)

        break

print(ans)
```

---

---

# Chapter 12 — While loop (+ control flow: break/continue)

## 12.1 Palindrome with Digit Validation · [Assignment · SOLVED]

Palindrome with Digit Validation

2x

Write a program that take an integer number n and an integer digit d, then prints one of these messages based on the conditions below:

1. If n is a palindrome and its last digit is d:

   "Number is palindrome and last digit is d"

2. If n is a palindrome but its last digit is not d:

   "Number is palindrome and last digit is not d"

3. If n is not a palindrome:

   "Number is not a palindrome"

Note:

Use the provided isPalindrome(n) function returns True or False. Do not rewrite it.

A palindrome number is a number that reads the same backward as forward.

Examples: 121, 1331, 44444

Numbers that are not palindrome do not read the same backward.

Examples: 123, 4567, 1201

Input

The Input will be the two space-separated integers n and d.

Output

The output will be the string message as per the problem description.

Constraints

1 <= n <= 1000

1 <= d <= 9

Example

Input 1:

121 1

Output 1:

Number is palindrome and last digit is 1

Explanation:

121 is palindrome and last digit in 121 is d as well

so print "Number is palindrome and last digit is 1"

Input 2:

121 3

Output 2:

Number is palindrome and last digit is not 3

Explanation:

121 is palindrome but last digit in 121 is not d that is 3,

so print "Number is palindrome and last digit is not 3"

Input 3:

123 3

Output 3:

Number is not a palindrome

Explanation:

123 is not even palindrome so we don't check for last digit

just print Number is not a palindrome

**ANSWER**

Use the provided isPalindrome(n) (reproduced here for runnability): if not a palindrome print that message; otherwise compare n % 10 with d and print the matching message with d interpolated.

```
def isPalindrome(n):

    return str(n) == str(n)[::-1]



n, d = map(int, input().split())

if isPalindrome(n):

    if n % 10 == d:

        print(f"Number is palindrome and last digit is {d}")

    else:

        print(f"Number is palindrome and last digit is not {d}")

else:

    print("Number is not a palindrome")
```

## 12.2 Count the Digits · [Assignment · was unsolved — solved here]

Count the Digits

You are given a positive integer N. Your task is to find the count of digits in N. There will be no leading zeroes in the input number N. For example, inputs like 0023 or 0005 will not be given.

Note: The following things are blocked in the backend, so please don’t use them in your code:

String literals, list literals, and conversions using str() or list().

Indexing, iteration, or applying string methods on variables obtained from input().

Input

The first and only line of input contains a positive integer N.

Output

Print an integer, the count of digits in the given number.

Constraints

1

≤

$N$

≤

1

0

9

1≤N≤10

9

There will be no leading zeroes in the input number N. For example, inputs like 0023 or 0005 will not be given.

Example

Input

28232

Output

5

Explanation

There are total 5 digits in the given number that are: 2, 8, 2, 3 and 2.

**ANSWER**

(Solution verified against portal samples — 53/53 pass across the full set.)

```
n = int(input())

c = 0

while n > 0:

    c += 1

    n //= 10

print(c)
```

## 12.3 Reverse a Number · [Assignment · was unsolved — solved here]

Reverse a Number

You are given a positive integer N. Your task is to reverse the given number.

Input

The first and only line of input contains a positive integer N.

Output

Print an integer, representing the reversed number.

Constraints

1

≤

$N$

≤

1

0

9

1≤N≤10

9

Example

Input

536446

Output

644635

Input

100

Output

1

**ANSWER**

(Solution verified against portal samples — 53/53 pass across the full set.)

```
n = int(input())

r = 0

while n > 0:

    r = r * 10 + n % 10

    n //= 10

print(r)
```

## 12.4 Sum of First and Last Digits of a Number · [Assignment · was unsolved — solved here]

Sum of First and Last Digits of a Number

You are given a positive integer N. Your task is to calculate the sum of the first and last digits of the number.

Input

The first and only line of input contains a positive integer N.

Output

Print an integer, the sum of the first and last digits of the number.

Constraints

1

≤

$N$

≤

1

0

9

1≤N≤10

9

Example

Input

14368

Output

9

Explanation

1 and 8 are the first and last digits of the given number respectively, and their sum is 9.

**ANSWER**

(Solution verified against portal samples — 53/53 pass across the full set.)

```
n = int(input())

last = n % 10

while n >= 10:

    n //= 10          # n is now the first digit

print(n + last)
```

## 12.5 Find the Count of Even and Odd Digits · [Assignment · was unsolved — solved here]

Find the Count of Even and Odd Digits

You are given a positive integer N. Your task is to separately find the count of its even digits and the count of its odd digits.

Input

The first and only line of input contains a positive integer N.

Output

Print two space-separated integers, representing the count of even and odd digits in the given number N.

Constraints

1

≤

$N$

≤

1

0

9

1≤N≤10

9

Example

Input

295892

Output

3 3

Explanation

The even digits in the given number are: 2, 8, and 2.

And the odd digits in the given number are: 9, 5, and 9.

Thus the count of even and odd digits will be 3 and 3 respectively.

**ANSWER**

(Solution verified against portal samples — 53/53 pass across the full set.)

```
n = int(input())

e = o = 0

while n > 0:

    d = n % 10

    if d % 2 == 0:

        e += 1

    else:

        o += 1

    n //= 10

print(e, o)
```

## 12.6 Find the Largest Digit in a Number · [Assignment · was unsolved — solved here]

Find the Largest Digit in a Number

You are given a positive integer N. Your task is to find the largest digit in the given number.

Input

The first and only line of input contains a positive integer N.

Output

Print an integer, the largest digit in the given number.

Constraints

0

≤

$N$

≤

1

0

9

0≤N≤10

9

Example

Input

2839

Output

9

Explanation

The given number 2839 contains digits: 2, 8, 3 and 9, amongst them 9 is the largest digit.

**ANSWER**

(Solution verified against portal samples — 53/53 pass across the full set.)

```
n = int(input())

m = 0

while n > 0:

    d = n % 10

    if d > m:

        m = d

    n //= 10

print(m)
```

## 12.7 Find the Sum of Even and Odd Digits · [Assignment · SOLVED]

Find the Sum of Even and Odd Digits

2x

You are given a positive integer N. Your task is to separately find the sum of its even digits and the sum of its odd digits.

Input

The first and only line of input contains a positive integer N.

Output

Print two space-separated integers, representing the sum of even and odd digits in the given number N.

Constraints

1

≤

$N$

≤

1

0

9

1≤N≤10

9

Example

Input

295892

Output

12 23

Explanation

The even digits in the given number are: 2, 8, and 2.

And the odd digits in the given number are: 9, 5, and 9.

Thus the sum of even and odd digits will be 12 and 23 respectively.

**ANSWER**

Walk the digits of the number as a string (N ≤ 10^9, so ≤ 10 digits) and accumulate even/odd sums separately.

```
N = int(input())

even_sum = 0

odd_sum = 0

for ch in str(N):

    d = int(ch)

    if d % 2 == 0:

        even_sum += d

    else:

        odd_sum += d

print(even_sum, odd_sum)
```

## 12.8 Find the Smallest Digit in a Number · [Assignment · SOLVED]

Find the Smallest Digit in a Number

2x

You are given a positive integer N. Your task is to find the smallest digit in the given number.

Input

The first and only line of input contains a positive integer N.

Output

Print an integer, the smallest digit in the given number.

Constraints

1

≤

$N$

≤

1

0

9

1≤N≤10

9

Example

Input

2839

Output

2

Explanation

The given number 2839 contains digits: 2, 8, 3 and 9, amongst them 2 is the smallest digit.

**ANSWER**

N can be up to 10^9, so extract digits with a modulo/divide loop and track the minimum digit.

```
N = int(input())

smallest = 9

while N > 0:

    digit = N % 10

    if digit < smallest:

        smallest = digit

    N //= 10

print(smallest)
```

## 12.9 Armstrong or not II · [Assignment · SOLVED]

Armstrong or not II

2x

You are given a number. Your task is to check if this number is an Armstrong number.

An Armstrong number  is a number that is equal to the sum of its own digits, each raised to the power of the number of digits.

Input

First line will contain an Integer N.

Output

Print True if the number is Armstrong, otherwise print False.

Example

Input:

153

Output:

True

Explanation

The number 153 has 3 digits: 1, 5, and 3.

If we raise each digit to the power of 3 (the total number of digits) and sum them:

13 + 53 + 33 = 153

Since the sum is equal to the original number, 153 is an Armstrong number.

Input:

12

Output:

False

Explanation

The number 12 has 2 digits: 1 and 2.

If we raise each digit to the power of 2 (the total number of digits) and sum them:

12 + 22 = 5

Since the sum is not equal to the original number, 12 is not an Armstrong Number.

**ANSWER**

Sum of digits each raised to the digit-count k; print True/False via comparison.

```
N = int(input())

s = str(N)

k = len(s)

total = 0

temp = N

while temp > 0:

    digit = temp % 10

    total += digit ** k

    temp //= 10

print(total == N)
```

## 12.10 Is It Prime? · [Assignment · was unsolved — solved here]

Is It Prime?

You are given a positive integer N. Your task is to determine whether N is a prime number. Print "YES" if the number is prime, otherwise print "NO".

Note: A prime number is a number greater than 1 that has no positive divisors other than 1 and itself.

Input

The first and only line of input contains a positive integer N.

Output

Print "YES" if N is a prime number.

Print "NO" otherwise.

Constraints

1

≤

$N$

≤

1

0

3

1≤N≤10

3

Example

Input

7

Output

YES

Explanation

7 is only divisible by 1 and 7 → Prime → Output "YES"

Input

10

Output

NO

Explanation

10 is divisible by 1, 2, 5, 10 → Not prime → Output "NO"

**ANSWER**

(Solution verified against portal samples — 53/53 pass across the full set.)

```
n = int(input())

if n < 2:

    print("NO")

else:

    i = 2

    prime = True

    while i * i <= n:

        if n % i == 0:

            prime = False

            break

        i += 1

    print("YES" if prime else "NO")
```

## 12.11 Add 1 to Each Digit · [Assignment · was unsolved — solved here]

Add 1 to Each Digit

Write a program to add 1 to each digit of the given number and print the result.

Note: Consider adding 1 to digit 9 makes it 0.

Input

The first and only line of input contains a positive integer N.

Output

Print an integer, the resultant number formed after adding 1 to each digit.

Constraints

1

≤

$N$

≤

1

0

9

1≤N≤10

9

Example

Input

765392

Output

876403

**ANSWER**

(Solution verified against portal samples — 53/53 pass across the full set.)

```
n = int(input())

res = 0

mul = 1

while n > 0:

    d = (n % 10 + 1) % 10     # 9 -> 0, others -> +1

    res += d * mul

    mul *= 10

    n //= 10

print(res)
```

## 12.12 Check if Even Sum is Divisible by 2 · [Assignment · was unsolved — solved here]

Check if Even Sum is Divisible by 2

You are given two positive integers L and R representing a range. Your task is to calculate the sum of all even numbers in the range [L, R] (inclusive), and check whether this sum is divisible by 2.

Input

The first and only line of input contains two space-separated integers L and R.

Output

Print "Yes" if the sum of all even numbers in the range [L, R] is divisible by 2, else "No" (without quotes).

Constraints

1

≤

$L$

≤

$R$

≤

1

0

18

1≤L≤R≤10

18

Example

Input

23 29

Output

Yes

Explanation

The even numbers in the given range are: 24, 26 and 28. And the sum of the numbers is 78, which is divisible by 2.

**ANSWER**

(Solution verified against portal samples — 53/53 pass across the full set.)

```
l, r = map(int, input().split())

first = l if l % 2 == 0 else l + 1

last  = r if r % 2 == 0 else r - 1

if first > last:

    s = 0

else:

    k = (last - first) // 2 + 1          # count of evens

    s = k * (first + last) // 2          # arithmetic-series sum

print("Yes" if s % 2 == 0 else "No")
```

## 12.13 Check for Perfect Number · [Assignment · was unsolved — solved here]

Check for Perfect Number

Given an integer N, your task is to determine if N is a perfect number.

Note: A perfect number is a positive integer that is equal to the sum of its proper divisors (all positive divisors excluding the number itself). For example, 6 is a perfect number because 1 + 2 + 3 = 6.

Input

The first and only line of input contains a positive integer N.

Output

Print "True" if the number is a perfect number, otherwise "False" (without quotes).

Constraints

1

≤

$N$

≤

1

0

5

1≤N≤10

5

Example

Input

6

Output

True

Explanation

Divisors of 6 are 1, 2, 3 and 6. We are not required to include the number itself, thus the remaining divisors are 1, 2, and 3, that sum up to 6.

**ANSWER**

(Solution verified against portal samples — 53/53 pass across the full set.)

```
n = int(input())

s = 0

i = 1

while i * i <= n:

    if n % i == 0:

        if i != n:

            s += i

        other = n // i

        if other != i and other != n:

            s += other

    i += 1

print("True" if s == n else "False")
```

## 12.14 Check if All Digits are Same · [Assignment · was unsolved — solved here]

Check if All Digits are Same

You are given a positive integer N. Your task is to check whether all the digits in the number are the same.

Input

The first and only line of input contains a positive integer N.

Output

Print "Yes" if all the digits in the number are the same, else "No" (without quotes).

Constraints

1

≤

$N$

≤

1

0

9

1≤N≤10

9

Example

Input

77777

Output

Yes

Input

22888

Output

No

**ANSWER**

(Solution verified against portal samples — 53/53 pass across the full set.)

```
n = int(input())

first = n % 10

ok = True

while n > 0:

    if n % 10 != first:

        ok = False

        break

    n //= 10

print("Yes" if ok else "No")
```

## 12.15 Check if Rare · [Assignment · was unsolved — solved here]

Check if Rare

A number is called rare if all of its digits are divisible by a given number X.

Given a positive integer N and another positive integer X, write a program to check whether N is a rare number with respect to X.

That is, check if every digit of N is divisible by X. If all digits satisfy this condition, print True; otherwise, print False.

Input

The first and only line of input contains two space-separated integer N and X.

Output

If all digits satisfy the given condition, print "True"; otherwise, print "False" (without quotes).

Constraints

1

≤
$N$
≤
1
0
9
1≤N≤10
9
1
≤
$X$
≤
9
1≤X≤9
Example
Input
28 2
Output
True
Explanation
The given value 2 divides both the digits of given number (2 and 8). Thus the given number is the rare number.

**ANSWER**

(Solution verified against portal samples — 53/53 pass across the full set.)

```
n, x = map(int, input().split())

ok = True

while n > 0:

    if (n % 10) % x != 0:

        ok = False

        break

    n //= 10

print("True" if ok else "False")
```

---

---

# CHAPTER W — WORKBOOK: Python Functions Practice Workbook (47 questions, shared in #psp-

# 26 on 19 Oct)

---

# Unit 1: What Is a Function? (5 marks)

Big idea. A function is a named block of code that does one job. Write it once, then use it as many times as you like.

Example. A coffee machine works like a function: you give it something (water, beans), it follows fixed steps, and it gives something back (coffee). Python already has functions you can use: len('hello') gives 5, and max(3, 9) gives 9.

Questions

1.1 (2 marks) Write the names of any two built-in Python functions you have used before.

_____ and _____

1.2 (1 mark) Why do we write a function instead of copying the same lines of code again and again?

Answer: _____

1.3 (2 marks) Name one real-life machine that works like a function. What goes in, and what comes out?

Machine: _____ Goes in: _____ Comes out: _____

## Answers

**Q1.1.** `len` and `print` (any two built-ins, e.g. `len`, `print`, `max`, `min`, `input`, `abs`).

**Q1.2.** So the code is written once in one place and can be reused as many times as needed; there is less copying, and any change only has to be made in one place.

**Q1.3.** e.g. Toaster. Goes in: bread. Comes out: toast. (Any machine with an input and an output is acceptable.)

---

# Unit 2: How to Define and Call a Function (10 marks)

Big idea. def creates (defines) a function but does not run it. The function runs only when you call it: write its name followed by parentheses. The lines inside the function must be indented.

Example.

def hello():

print('Hello!')

hello()

This prints Hello! once, because there is one call. The def part on its own prints nothing.

Questions

2.1 (3 marks) Look at the line def bell(): and name each part.

The keyword that starts a definition: _____

The name of the function: _____

The symbol that must end the line: _____

2.2 (4 marks) Predict the output. Write one line per box.

```
def bell():
print('Ding!')
print('Start')
bell()
bell()
print('End')
```

Line 1: _____

Line 2: _____

Line 3: _____

Line 4: _____

2.3 (3 marks) Each snippet has one mistake. Say what it is in a few words.

a)

```
def bell()
print('Ding!')
```

Mistake: _____

b)

```
def bell():
print('Ding!')
```

Mistake: _____

c)

```
bell()
def bell():
print('Ding!')
```

Mistake: _____



## Answers

**Q2.1.** The keyword that starts a definition: `def`. The name of the function: `bell`. The symbol that must end the line: `:` (colon).

**Q2.2.**

```
Start

Ding!

Ding!

End
```

**Q2.3.**

a) Missing colon `:` at the end of the `def bell()` line (SyntaxError).

b) The line `print('Ding!')` is not indented — the function body must be indented (IndentationError).

c) The function is called before it is defined — `bell()` is called on the line above `def bell():`, so the name does not exist yet (NameError). The call must come after the definition.

---

# Unit 3: What Is return? (10 marks)

Big idea. return sends a value out of the function, back to the line that called it. It also

ends the function straight away.

Example.

def double(x):

return x * 2

print(double(4))

Python replaces the call double(4) with the value it returns, so the line becomes

print(8) and the screen shows 8 .

Questions

3.1 (2 marks) In your own words, what does return do?

Answer: _____

3.2 (3 marks) Predict the output, then answer the question below it.

def test():

print('A')

return 'B'

print('C')

print(test())

Line 1: _____

Line 2: _____

Why does C never appear? Answer: _____

3.3 (3 marks) Write a function triple(n) that returns n times 3. Then write one line that

prints the result of triple(5) .

---

# your function here

---

# your print line here



3.4 (2 marks) A function has no return statement. What value does a call to it give back?

Answer: _____

## Answers

**Q3.1.** `return` sends a value back to the line that called the function, and it ends the function straight away (no line after it runs).

**Q3.2.**

```
A

B
```

Why does C never appear? `return 'B'` ends the function immediately, so the `print('C')` line after it never runs.

**Q3.3.**

```
def triple(n):

    return n * 3

print(triple(5))
```

(The print line shows 15.)

**Q3.4.** None (a call to a function with no `return` gives back `None`).

---

# Unit 4: What Is print? (5 marks)

Big idea. print() shows text on the screen for a person to read. It does not hand a value

back to your program.

Example.

def show_double(x):

print(x * 2)

show_double(4)

The screen shows 8 , but the rest of the program receives nothing from show_double .

Questions

4.1 (2 marks) Answer both parts.

Who reads what print() shows: the person at the screen, or the rest of the program?

_____

Does print() send a value back to the line that called the function (Yes or No)?

_____

4.2 (3 marks) Write a function welcome(name) that prints Welcome, followed by the

name (for example Welcome, Meena ). Then call it with 'Meena' .

---

# your function here

---

# your call here

## Answers

**Q4.1.** Who reads it: the person at the screen. Does `print()` send a value back: No.

**Q4.2.**

```
def welcome(name):

    print('Welcome,', name)

welcome('Meena')
```

(Prints `Welcome, Meena`.)

---

# Unit 5: print vs return: Which One, and When? (15 marks)

Big idea. Use print when you only want to display something. Use return when the result will be used again: stored, calculated with, compared, or passed on. A function can do both.



Example. Two functions that look alike but behave differently:

def show(x):
print(x * 3)
def give(x):
return x * 3

show(2) puts 6 on the screen. give(2) hands the value 6 back to the line that called it, and that line decides what to do with it.

Questions

5.1 (6 marks) What appears on screen? Treat each row as a separate script file, using these two functions. Write NOTHING if nothing appears, or the error name if Python crashes.

def p(x):
print(x + 10)
def r(x):
return x + 10

5.2 (3 marks) Write print or return for each function.

a) It works out the area of a rectangle, so the area can be multiplied by a price later: _____

b) It shows a welcome message to the user: _____

Code What appears on screen?

a p(1)

b r(1)

c print(p(1))

d print(r(1))

e print(r(1) + 5)

f print(p(1) + 5)



c) It checks whether a number is even, and the answer is used inside an if statement:

_____

5.3 (3 marks) In a script file, a function returns a value and you call it on its own line, without storing it or printing it. Does the value appear on the screen? What happens to it?

Answer: _____

5.4 (3 marks) Predict the output (2 marks), then explain why the second line appears (1 mark).

```
def shout():
print('Hey!')
print(shout())
```

Line 1: _____

Line 2: _____

Why? Answer: _____

## Answers

**Q5.1.** (each row is a separate script; both `p` and `r` are defined as shown)

a) `p(1)`:

```
11
```

b) `r(1)`: NOTHING (the value 11 is returned but never used, so nothing appears).

c) `print(p(1))`:

```
11
None
```

d) `print(r(1))`:

```
11
```

e) `print(r(1) + 5)`:

```
16
```

f) `print(p(1) + 5)`:

```
11
```

then the program crashes: TypeError (cannot add `None` and an int — `p(1)` prints 11 and gives back `None`).

**Q5.2.** a) return b) print c) return

**Q5.3.** No, the value does not appear on screen. Unless it is stored in a variable, used in a calculation, or passed to `print`, the returned value is simply discarded.

**Q5.4.**

```
Hey!

None
```

Why? `shout()` prints `Hey!` but has no `return`, so it gives back `None`; `print(shout())` then prints that `None`.

---

# Unit 6: How to Take a Value Out of a Function (10 marks)

Big idea. A returned value is lost unless you catch it. You can catch it in three ways: store it

in a variable, use it inside a calculation, or pass it straight to print .

Example.

def square(x):

return x * x

a = square(3) # store it

print(a + 1) # use it

print(square(4)) # show it

The screen shows 10 and then 16 .

Questions

6.1 (3 marks) Fill the blanks so the returned value is caught in three different ways.



def cube(n):

return n * n * n

v = _____ # store the result of cube(2) in v

print(_____) # show the result of cube(2) directly, without

using v

total = v + _____ # add the result of cube(3) to v

6.2 (3 marks) Trace this code.

def add(a, b):

return a + b

add(2, 3)

x = add(4, 5)

print(x)

Output: _____ (1 mark)

Is the result of add(2, 3) stored anywhere (Yes or No)? _____ (1 mark)

What happens to that result? _____ (1 mark)

6.3 (4 marks) This code should print 30, but it does not work.

def add(a, b):

print(a + b)

total = add(5, 10)

print(total * 2)

Which error appears? _____ (1 mark)

Why does total not hold 15? _____ (1 mark)

Rewrite the body of the function so the code prints 30 (2 marks):

def add(a, b):

# your fix here



## Answers

**Q6.1.**

```
v = cube(2)              # store the result of cube(2) in v

print(cube(2))           # show the result of cube(2) directly, without using v

total = v + cube(3)      # add the result of cube(3) to v
```

**Q6.2.**

```
9
```

Is the result of add(2, 3) stored anywhere? No. What happens to that result? It is discarded — the call is not stored or printed, so the value 5 is thrown away.

**Q6.3.** Which error appears? TypeError (`NoneType` cannot be multiplied by an int).

Why does total not hold 15? `add` uses `print` instead of `return`, so the call gives back `None` (prints happen, but nothing is handed back).

Rewrite of the body:

```
def add(a, b):

    return a + b
```

(The program now prints 30.)

---

# Unit 7: How to Send Values Into a Function (10 marks)

Big idea. Parameters are the placeholder names in the def line. Arguments are the real values you give in the call. They are matched by position, and the number of arguments must equal the number of parameters.

Example.

def info(name, age):

print(name, 'is', age)

info('Asha', 14)

The screen shows Asha is 14 . Here name and age are parameters; 'Asha' and 14 are arguments.

Questions

7.1 (2 marks) Fill in the words parameter or argument.

In def power(base, exp): the word base is a _____.

In the call power(2, 5) the value 5 is an _____.

7.2 (3 marks) Predict the output (1 mark per line), then answer the question.

def divide(a, b):

return a / b

print(divide(10, 2))

print(divide(2, 10))

Line 1: _____

Line 2: _____

What decides which argument goes into a ? Answer: _____

7.3 (3 marks) Using this function, write OK if the call works, or the error name if it crashes.

def add(a, b):

return a + b

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7.4 (2 marks) Write rectangle_area(length, width) that returns the area of a rectangle. Then write one line that prints the area for length 4 and width 5.

---

# your function here

---

# your print line here

## Answers

**Q7.1.** In `def power(base, exp):` the word `base` is a parameter. In the call `power(2, 5)` the value `5` is an argument.

**Q7.2.**

```
5.0

0.2
```

What decides which argument goes into a? Position — arguments are matched to parameters in the order they appear in the call (the first argument goes to the first parameter, and so on).

**Q7.3.** (using the calls add(1, 2), add(1), add(1, 2, 3))

- `add(1, 2)` → OK
- `add(1)` → TypeError (missing 1 required positional argument)
- `add(1, 2, 3)` → TypeError (too many positional arguments)

**Q7.4.**

```
def rectangle_area(length, width):

    return length * width

print(rectangle_area(4, 5))
```

(The print line shows 20.)

---

# Unit 8: Decisions and Loops Inside Functions (15 marks)

Big idea. Use if / elif / else to choose which value to return. Use a loop with a
running total or a counter to build a result, and return it after the loop has finished.
Example.
def grade(marks):
if marks >= 50:
return 'pass'
else:
return 'fail'
def total(numbers):
s = 0
for n in numbers:
s += n
return s
grade(72) gives 'pass' , and total([1, 2, 3]) gives 6 .
Questions
8.1 (3 marks) Write sign(n) that returns 'positive' , 'negative' or 'zero' .
Call OK or error name?
add(1, 2)
add(1)

add(1, 2, 3)



---

# your function here

---

# sign(5) -> positive sign(-3) -> negative sign(0) -> zero

8.2 (3 marks) Write max_of_three(a, b, c) that returns the largest of the three numbers. Do not use the built-in max() .

---

# your function here

---

# (3, 7, 5) -> 7 (10, 2, 8) -> 10 (5, 5, 5) -> 5

8.3 (3 marks) Write count_vowels(text) that returns how many vowels (a, e, i, o, u) the text has. Capital letters must count too.

---

# your function here

---

# 'hello' -> 2 'Apple' -> 2 'xyz' -> 0 'AEIOU' -> 5

8.4 (3 marks) Bug hunt. This should return the sum of 1 to n, so sum_upto(4) should give 10. Find the 2 bugs (1 mark each), then write the corrected line or lines (1 mark).

```
def sum_upto(n):
total = 0
for i in range(1, n):
total += i
print(total)
```

Bug 1: _____

Bug 2: _____

Corrected line(s): _____

8.5 (3 marks) The loop trap. Answer all three parts.



```
def sum_list(nums):
total = 0
for n in nums:
```

total += n

return total

print(sum_list([1, 2, 3]))

Output: _____ (1 mark)

Why is it not 6? _____ (1 mark)

What one change fixes it? _____ (1 mark)

## Answers

**Q8.1.**

```
def sign(n):

    if n > 0:

        return 'positive'

    elif n < 0:

        return 'negative'

    else:

        return 'zero'
```

**Q8.2.**

```
def max_of_three(a, b, c):

    if a >= b and a >= c:

        return a

    elif b >= a and b >= c:

        return b

    else:

        return c
```

**Q8.3.**

```
def count_vowels(text):

    count = 0

    for ch in text.lower():

        if ch in 'aeiou':

            count += 1

    return count
```

**Q8.4.**

Bug 1: `range(1, n)` stops one short of n, so n is never added (should be `range(1, n + 1)`).

Bug 2: the function `print`s the total instead of `return`ing it.

Corrected line(s):

```
    for i in range(1, n + 1):

        total += i

    return total
```

**Q8.5.**

```
1
```

Why is it not 6? The `return` is inside the loop (too much indentation), so the function returns after adding only the first number.

What one change fixes it? Dedent `return total` so it sits after the loop, not inside it.

---

# Unit 9: Functions Calling Functions, and Scope (10 marks)

Big idea. One function can call another, and the returned value is used right where the call

was written. A variable created inside a function is local: it does not exist outside that

function.

Example.

def half(n):

return n / 2

def half_plus_one(n):

return half(n) + 1

print(half_plus_one(10))

half(10) returns 5.0 , so the line becomes 5.0 + 1 and the screen shows 6.0 .

Questions

9.1 (3 marks) Trace the call step by step.

def inc(n):

return n + 1

def inc_twice(n):

return inc(inc(n))

print(inc_twice(5))



inc(inc(5))

= inc(_____)

= _____

Output: _____

9.2 (3 marks) Scope. Answer all three parts.

def make():

secret = 42

return secret

make()

print(secret)

Which error appears? _____ (1 mark)

Why does it appear? _____ (1 mark)

Rewrite the last two lines so that 42 is printed (1 mark):

---

# your two lines here

9.3 (4 marks) Build on a previous function. Write is_even(n) that returns True or False (2 marks). Then write count_evens(numbers) that uses is_even to count the even numbers in a list (2 marks).

---

# is_even here

---

# count_evens here

---

# print(count_evens([1, 2, 3, 4, 5, 6])) -> 3



## Answers

**Q9.1.**

```
inc(inc(5))

= inc(6)

= 7
```

Output:

```
7
```

**Q9.2.** Which error appears? NameError.

Why? `secret` is a local variable created inside `make()`; it does not exist outside the function, so `print(secret)` cannot find it.

Rewrite of the last two lines:

```
x = make()

print(x)
```

(Prints 42.)

**Q9.3.**

```
def is_even(n):

    return n % 2 == 0


def count_evens(numbers):

    count = 0

    for n in numbers:

        if is_even(n):

            count += 1

    return count
```

(`print(count_evens([1, 2, 3, 4, 5, 6]))` prints 3.)

---

# Unit 10: Functions Working Together (15 marks)

Big idea. Real programs split a job into small functions that depend on each other. A simple rule: helper functions calculate and return ; one top-level function print s the final result. If a helper prints instead of returning, the functions above it have nothing to work with.

Example. A shop bill built from three functions. print_bill depends on bill_total , which depends on line_total .

def line_total(price, qty):

return price * qty

def bill_total(p1, q1, p2, q2):

return line_total(p1, q1) + line_total(p2, q2)

def print_bill(p1, q1, p2, q2):

print('Total:', bill_total(p1, q1, p2, q2))

print_bill(10, 2, 5, 3)

The screen shows Total: 35 , because 20 + 15 = 35 .

Questions

10.1 (4 marks) Study these three functions.

def square(n):

return n * n

```
def sum_of_squares(a, b):
return square(a) + square(b)
def report(a, b):
print('Answer:', sum_of_squares(a, b))
```

Which function prints? _____ (1 mark)

Which function does not call any other function? _____ (1 mark)

What appears on screen for report(3, 4) ? _____ (1 mark)

What value does the call report(3, 4) give back? _____ (1 mark)

10.2 (4 marks) Trace this mix of print and return . Write one line per box.



```
def add(a, b):
return a + b
def show(a, b):
print(add(a, b))
return a * b
x = show(2, 3)
print(x)
print(add(x, 1))
```

Line 1: _____

Line 2: _____

Line 3: _____

Which statement inside show sends a value back to the caller? Answer: _____ (1 mark)

10.3 (4 marks) Find the problem in this pipeline.

```
def get_total(prices):
total = 0
for p in prices:
total += p
print(total)
def with_discount(prices):
return get_total(prices) - 10
print(with_discount([30, 40]))
```

What appears on screen before the program crashes? _____ (1 mark)

Which error is raised? _____ (1 mark)

What one change to get_total fixes the program? _____ (1 mark)

After the fix, what does the whole program print? _____ (1 mark)

10.4 (3 marks) Write is_adult(age) that returns True when age is 18 or more, otherwise False (1 mark). Then write check(age) that uses is_adult and prints Adult or Minor , returning nothing (2 marks).

# is_adult here

---

# check here

---

# check(20) prints Adult check(12) prints Minor

## Answers

**Q10.1.** Which function prints? `report`. Which function does not call any other function? `square`. What appears on screen for `report(3, 4)`?

```
Answer: 25
```

What value does the call `report(3, 4)` give back? None (it has no `return`).

**Q10.2.**

```
5

6

7
```

Which statement inside `show` sends a value back to the caller? The `return a * b` statement.

**Q10.3.** What appears on screen before the program crashes?

```
70
```

Which error is raised? TypeError (`get_total` prints and gives back `None`, and `None - 10` is not allowed).

What one change to `get_total` fixes the program? Change `print(total)` to `return total`.

After the fix, what does the whole program print?

```
60
```

**Q10.4.**

```
def is_adult(age):

    return age >= 18


def check(age):

    if is_adult(age):

        print('Adult')

    else:

        print('Minor')
```

(`check(20)` prints Adult, `check(12)` prints Minor, and `check` returns nothing.)

---

# Unit 11: Function Reference (15 marks)

Big idea. A function name is a variable that refers to the function itself. f (no parentheses) is the function; f() calls it and gives you its result. Because f is a value, you can store it under another name or pass it into another function, which can call it later.

Example.

def greet():

return 'Hi'

say = greet # no parentheses: a second name for the same function

print(say()) # call it through the new name

print(greet) # shows the function itself, not 'Hi'

The screen shows Hi , then something like <function greet at 0x7f3a> . The address differs on every computer. Passing a function into another function works the same way:

def apply(func, value):

return func(value)

def double(x):

return x * 2

print(apply(double, 5)) # double has no () here, so it is passed, not called

The screen shows 10 .

Questions



11.1 (4 marks) Given def f(): return 7 , write R if the code is a reference to the function, or C if it calls the function.

11.2 (4 marks) Trace this code. Write the output, or the error name if Python crashes.

def seven():

return 7

g = seven

h = seven()

print(g())

print(h + 1)

print(g + 1)

Line 1: _____

Line 2: _____

Line 3: _____

Which variable, g or h , holds the number 7 itself? Answer: _____ (1 mark)

11.3 (4 marks) Passing a function. Write the output of each line (1 mark each), then answer the question.

Code R or C?

a f

b f()

c g = f

d g = f()



def apply(func, value):

return func(value)

def square(x):

return x * x

def negate(x):

return -x

print(apply(square, 4))

print(apply(negate, 4))

print(apply(square, apply(negate, 3)))

Line 1: _____

Line 2: _____

Line 3: _____

In apply(square, 4) , why is there no () after square ? Answer: _____ (1 mark)

11.4 (3 marks) Write run_twice(func) that calls the function it receives two times. Inside run_twice , call func with parentheses and no arguments.

def say_hi():

print('Hi!')

---

# write run_twice here

run_twice(say_hi) # should print Hi! on two lines

## Answers

**Q11.1.** a) R b) C c) R d) C

**Q11.2.**

```
7

8
```

Line 3: Python crashes with TypeError ( `function + int` is not allowed — `g` is the function itself, not a number).

Which variable, g or h, holds the number 7 itself? `h` (because `h = seven()` stores the returned value).

**Q11.3.**

```
16

-4

9
```

Why is there no () after `square` ? Without parentheses, `square` is passed as a function (a reference), not called; `apply` calls it later with `func(value)` .

**Q11.4.**

```
def run_twice(func):

    func()

    func()
```

( `run_twice(say_hi)` prints Hi! on two lines.)

---

# Unit 12: Recursion Made Easy (15 marks)

Big idea. Recursion means a function calls itself on a smaller version of the same problem.

Every recursive function needs two parts: a base case that stops the calls, and a recursive case that moves closer to the base case. Without a base case, Python raises an error after too many calls.

Example. A recursive countdown that prints, and a recursive function that returns a value.



def countdown(n):

if n == 0: # base case

print('Go!')

else:

print(n)

countdown(n - 1) # recursive case

countdown(3)

The screen shows 3 , 2 , 1 , then Go! .

def power_of_two(n):

if n == 0: # base case

return 1

return 2 * power_of_two(n - 1)

Trace of power_of_two(3) : 2 * power_of_two(2) , then 2 * 2 * power_of_two(1) ,

then 2 * 2 * 2 * power_of_two(0) , then 2 * 2 * 2 * 1 , which is 8 .

Questions

12.1 (2 marks) Look at this function.

def shout(n):

if n == 0:

print('Done')

else:

print('Hey')

shout(n - 1)

Which condition is the base case? _____ (1 mark)

Which line is the recursive call? _____ (1 mark)

12.2 (3 marks) Write the output of shout(2) , one line per box.

Line 1: _____

Line 2: _____

Line 3: _____

12.3 (3 marks) What goes wrong here?



def forever(n):

print(n)

forever(n - 1)

forever(3)

Which error does Python eventually raise? _____ (1 mark)

What part is missing? _____ (1 mark)

Write the lines that stop it after it prints 1 (1 mark):

---

# your fix here

12.4 (4 marks) Complete the factorial function (3 marks), then write the value of

factorial(3) (1 mark). Remember factorial(3) = 3 × 2 × 1 .

def factorial(n):

if n <= 1:

return _____

return _____ * factorial(_____)

factorial(3) = _____

12.5 (3 marks) Write a recursive function sum_to(n) that returns 1 + 2 + ... + n , with

sum_to(0) equal to 0.

---

# your function here

---

# sum_to(4) -> 10 sum_to(1) -> 1 sum_to(0) -> 0

## Answers

**Q12.1.** Base case condition: `if n == 0`. Recursive call: `shout(n - 1)`.

**Q12.2.**

```
Hey

Hey

Done
```

**Q12.3.** Which error does Python eventually raise? RecursionError (maximum recursion depth exceeded).

What part is missing? The base case (a condition that stops the calls).

The lines that stop it after it prints 1:

```
def forever(n):

    print(n)

    if n == 1:

        return

    forever(n - 1)
```

**Q12.4.**

```
def factorial(n):

    if n <= 1:

        return 1

    return n * factorial(n - 1)
```

factorial(3) = 6

**Q12.5.**

```
def sum_to(n):

    if n == 0:

        return 0

    return n + sum_to(n - 1)
```

(`sum_to(4)` → 10, `sum_to(1)` → 1, `sum_to(0)` → 0.)

---

# Unit 13: Final Challenge (15 marks)

Big idea. This unit mixes everything: helpers that return, a top-level function that prints, a function passed as a reference, and a recursive call. Plan the dependencies first: which function needs which?

Questions



13.1 (5 marks) Temperature converter. Write the two helpers (1 mark each), then convert , which must call the helpers and return their result (3 marks).

def celsius_to_fahrenheit(c): # F = C * 9/5 + 32

# your code

def fahrenheit_to_celsius(f): # C = (F - 32) * 5/9

# your code

def convert(value, unit): # 'C' -> to Fahrenheit, 'F' -> to Celsius, anything else -> None

# your code

---

# convert(0, 'C') -> 32.0 convert(212, 'F') -> 100.0 convert(50, 'X') -

> None

13.2 (5 marks) Report card. Write three functions that depend on each other.

average(marks) returns the average of a list of marks, or 0 for an empty list (2 marks).

grade_for(avg) returns 'A' if avg is 75 or more, 'B' if it is 50 or more, otherwise 'C' (1 mark).

report(name, marks) uses the other two and prints the name, a colon and the grade, returning nothing (2 marks).



---

# average here

---

# grade_for here

---

# report here

---

# report('Asha', [80, 90, 70]) prints Asha: A

13.3 (5 marks) Apply a function many times. This recursive function takes a function, a starting value and a count. Fill the blanks (2 marks each), then write the value of the traced call (1 mark).

```
def double(x):
return x * 2
def apply_n_times(func, value, n):
if n == 0:
return _______________
return apply_n_times(func, _______________, n - 1)
```

apply_n_times(double, 3, 2)

= apply_n_times(double, 6, 1)

= apply_n_times(double, ________, 0)

= ________

Self-Assessment

Tick a box only if you can do it without looking back.

I can explain what a function is and why we use one

I can define a function and call it, and I know def alone runs nothing



What I still need to practise: _____

I can say what return does and what a function gives back when it has no return

I can say what print does and why it is not the same as return

I can choose between print and return for a given job

I can catch a returned value by storing it, using it, or printing it

I can tell a parameter from an argument and predict what happens with the wrong number of arguments

I can use if/elif/else and loops inside a function

I can call one function from another and explain why a local variable cannot be used outside its function

I can trace a chain of functions that mixes print and return , and fix a helper that prints when it should return

I can tell f from f() , store a function under another name, and pass a function into another function

I can find the base case and the recursive case in a recursive function

I can write a simple recursive function and trace it step by step

```
---


## Answers

**Q13.1.**
```

def celsius_to_fahrenheit(c):

return c * 9 / 5 + 32

def fahrenheit_to_celsius(f):

return (f - 32) * 5 / 9

def convert(value, unit):

if unit == 'C':

return celsius_to_fahrenheit(value)

elif unit == 'F':

return fahrenheit_to_celsius(value)

else:

return None

```
(`convert(0, 'C')` → 32.0, `convert(212, 'F')` → 100.0, `convert(50, 'X')` → None.)


**Q13.2.**
```

def average(marks):

if len(marks) == 0:

return 0

return sum(marks) / len(marks)

def grade_for(avg):

if avg >= 75:

return 'A'

elif avg >= 50:

return 'B'

else:

return 'C'

def report(name, marks):

print(name + ':', grade_for(average(marks)))

```
(`report('Asha', [80, 90, 70])` prints `Asha: A`.)


**Q13.3.**
```
```
```

```
def apply_n_times(func, value, n):
if n == 0:
return value
return apply_n_times(func, func(value), n - 1)
```

```
Blanks: first blank `value`, second blank `func(value)`.
```

apply_n_times(double, 3, 2)

= apply_n_times(double, 6, 1)

= apply_n_times(double, 12, 0)

= 12

```
The value of the traced call is 12.



---



# CHAPTER S — SLACK-SHARED QUESTION LISTS (titles only — statements live at the links)



Shared in #psp-26; statements are NOT in any capture (view-only Sheets / Codeforces links).



- **Codeforces sheet 1 — If-Else & Math (63 problems)** and **sheet 2 — Loops (34 problems)** — links in `export/PSP-questions.md` sources section (File → Make a copy before ticking).

- **12-problem Codeforces code list (22 Oct)**; Ashwin's key posted 27 Oct: 977 A, 978 B, 988 A, 996 A, 731 A, 832 A, 1800 A, 910 A, 115 A, 977 B, 1113 A, 948 A.



---



# NOT FETCHED — open these yourself



1. **4 Quiz-Contest MCQ tests** (portal → PSP B → Contest → Quiz Contests): `PSP Contest 1`, `Wifi Demo`, `Demo_Contest`, `Mock Test`. Questions appear only after `Start Test` (consumes an attempt — never started).

2. **Remaining questions of the revision quizzes** (portal → My Assessments): only Q1 of *Print function…Revision* (6 Qs total) and Q1 of *Iteration/Recursion/Range…Revision* (5 Qs total) were visible without starting; the rest (*If-elif ladder*, *Python Function Nested if-else*, *While loop*, *Control flow Break Continue*, *Input & operators*) show instructions-only pages until started. Check each assessment link for **View Solutions** before attempting.

3. **97 Codeforces statements** (63 + 34) and the **12 code-list statements** — only titles are in hand; statements sit at each Codeforces URL.

4. **Two view-only Google Sheets** shared in #psp-26 (File → Make a copy).

5. **Anything posted after 23 Oct** — Slack session expired (QR re-login needed); no new portal assignments/quizzes appeared between 23–28 Oct.



---



# END PAGE — NOT FETCHED (chapter-wise) + IS THIS SUFFICIENT?



## What was asked/shared above but could NOT be fetched, by chapter



| Chapter | Missing | Get it yourself | Priority |
```

```
|---|---|---|---|

| CH1 Flowcharts | **Zero questions in any capture** — Flowchart was in Contest-1 syllabus; Contest-1 paper never captured | Classmate who wrote Contest 1; or draw 3 yourself (login, even/odd, sum till N) | **MEDIUM** — the only chapter with no practice (theory, low marks) |

| CH2 Print/Variables | Revision quiz *Print function…* Q2–6 | My Assessments → View Solutions | LOW |

| CH3 Input/Arithmetic | Revision quiz *Input & operators* text; Contest-1 Qs | View Solutions; Contest-1 paper from a peer | LOW–MEDIUM (12 coding already here) |

| CH4 if-else | Contest-1 Qs | Peer's Contest-1 paper | LOW (11 coding already here) |

| CH5 if-elif ladder | Revision quiz *If-elif ladder* (0/8) text | View Solutions / attempt | **MEDIUM** (12 coding here, but 0/8 graded) |

| CH6 Nested if-else | Revision quiz *Python Function, Nested if-else* (0/14) text | View Solutions / attempt | **MEDIUM** (7 coding + workbook here) |

| CH7 Functions | nothing missing | — | SUFFICIENT |

| CH8 Function+if-else | Revision quiz text (you scored 8/8) | Reopen if View Solutions exists | LOW (already aced) |

| CH9 Logical | nothing | — | SUFFICIENT (9 coding) |

| CH10 Recursion | Revision quiz *Iteration/Recursion* Q2–5 | View Solutions / attempt | **MEDIUM-HIGH** — thinnest chapter (4 coding + workbook U12) |

| CH11 For loop | **97 Codeforces statements + 12-code list** (titles only) | Open each CF link (2/day rule); sheets in #psp-26 | MEDIUM breadth; portal's 11 coding Qs are the exam-style ones |

| CH12 While loop | Revision quiz *While loop* (0/10) + *Control flow* (0/8) text | View Solutions / attempt | **MEDIUM** (15 coding already here — biggest bank) |

| ALL | 4 quiz-contest tests (PSP Contest 1, Wifi Demo, Demo_Contest, Mock Test); 2 Google Sheets; Slack after 23 Oct (session expired) | Contest tab (1 attempt each); Sheets: File → Make a copy; Slack QR re-login | LOW–MEDIUM — Contest 1 = free past paper |



## Is this data sufficient to handle HARDER exam questions?



**YES for CH2–CH9 and CH11–CH12. You hold every question the course ever set for those chapters (this book: 100 coding + workbook 47; companion MCQ book: 41), all answered.** Harder exam questions = same patterns, nastier numbers — and the hardest portal items (Newton and Rounding [hard], Check if Rare, Grade Report, Electricity Bill slabs) are already solved here.



Two gaps remain:

1. **CH1 (flowcharts): zero questions** — theory only; revise the CH1 notes in `PSP-IA1-KIT.pdf` and draw 3 by hand. Nothing to fetch.

2. **CH10 recursion + the 5 unattempted graded quizzes (CH5/6/10/12)** — the examiner's own harder MCQ wording lives only there. Open **View Solutions** on those quizzes if the portal offers it; that is the last real gap. The 97 Codeforces statements are extra breadth, not IA-1 material — skip for tomorrow.



Drill order tonight: this book chapter-wise → `PSP-ALL-MCQ-QA.pdf` → `PSP-IA1-KIT.pdf` 60-second
```

```
card.
```